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VALIDATION REPORT

LAND SURFACE TEMPERATURE

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List of Acronyms

ABI	Advanced Baseline Imager
AD	Applicable Document
AHI	Advanced Himawari Imager
ATBD	Algorithm Theoretical Basis Document
C3S	Copernicus Climate Change Service
CAL/VAL	Calibration / Validation
CCI	(ESA) Climate Change Initiative
CEOS	Committee on Earth Observation Satellites
CGLOPS	Copernicus Global Land Operations
CLMS	Copernicus Land Monitoring Service
COG	Cloud Optimized Geotiff
CSV	Comma-Separated Values
DC	Diurnal Cycle
ECMWF	European Centre for Medium-Range Weather Forecasts
ECV	Essential Climate Variable
EUMETSAT	European Agency for the Exploitation of Meteorological Satellites
ESA	European Space Agency
FCOVER	Fraction of Green Vegetation Cover
GBOV	Copernicus Ground-Based Observations for Validation
GCOS	Global Climate Observing System
GEO	Geostationary
GOES	Geostationary Operational Environmental Satellite
GSW	Generalized Split-Windows
HMWR	Himawari
IODC	Indian Ocean Data Coverage mission
IMS	Snow and Ice Mapping System
INSPIRE	Infrastructure for Spatial Information in the European Community
IPMA	Instituto Português do Mar e da Atmosfera
JASMIN	Joint Analysis System Meeting Infrastructure Needs
JMA	Japan Meteorological Agency
JRC	Joint Research Council
LSA-SAF	Satellite Application Facility on Land Surface Analysis
LST	Land Surface Temperature
LST10	Land Surface Temperature-10 daily composite
LPV	Land Product Validation
Max	Maximum
Med	Median
MIR	Middle InfraRed
Min	Minimum
MSG	Meteosat Second Generation
MTG	Meteosat Third Generation

MTSAT	Multi-Functional Transport Satellite
NetCDF	Network Common Data Format
NHEMI	Northern Hemisphere
NOAA	National Oceanic and Atmospheric Administration
NRT	Near-Real Time
NWC-SAF	Satellite Application Facility on Support to Nowcasting and Very Short-Range Forecasting
PQAD	Product Quality Assurance Document
PROBA-V	Project for On-Board Autonomy - Vegetation
PUM	Product User Manual
RM	Reference Measurement
QAR	Quality Assessment Report
RMSD	Root Mean-Squared Difference
S2GM	Sentinel-2 Global Mosaic
SAF	Satellite Application Facility
SCE	Snow Cover Extent
SKT	Skin Temperature
SLSTR	Sea and Land Surface Temperature Radiometer
SLSTR-A	(Sentinel-3A) Sea and Land Surface Temperature Radiometer
SYN	Synthesis
TCI	Thermal Condition Index
TIR	Thermal InfraRed
UTC	Universal Time Coordinated
V_{PREV}	Previous CLMS LST Operational Product Version (V1.0–V2.2)
VR	Validation Report
WGS	World Geodetic System

EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time-series.

Quality Assessment, and continuous Quality Monitoring constitute the only means of guaranteeing the compliance of generated products with requirements:

- The former concerns the new products/version which must pass an exhaustive scientific evaluation before to be implemented operationally.
- The latter concerns the operational products to check their quality keeps the same level along the time.

Both follow, as much as possible, the guidelines, protocols and metrics defined by the Land Product Validation (LPV) group of the Committee on Earth Observation Satellite (CEOS) for the validation of satellite-derived land products.

This report presents the validation results of the CLMS Land Surface Temperature (LST) product version 3.0 (V3.0), and assesses the impact of the successive updates implemented in the previous operational version (from V1.0 to V2.2, hereafter collectively referred to as V_{PREV}). The CLMS LST product provides combined LST estimates derived from a constellation of geostationary (GEO) satellite missions: Meteosat Second Generation (MSG) – 0° service and Indian Ocean Data Coverage (IODC) service; Geostationary Operational Environmental Satellite (GOES); and Himawari (HMWR). Although these platforms are operated by different international satellite agencies, the respective data are made available in near-real time through the European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT) dissemination system, EUMETCast.

The product incorporates LST estimates for the MSG 0º mission provided by the Land Surface Analysis Satellite Application Facility (LSA-SAF). The LSA-SAF algorithm and associated products enable the use of dynamic vegetation cover to estimate surface emissivity. Within the CLMS, the availability of a comparable global dataset (the FCOVER 1 km product derived from PROBA-V observations) allows the application of a similar emissivity estimation approach to the sensors processed by the CLMS processing chain (i.e. GOES-16 and HMWR-8/9). The upgrade to V3.0 introduces the following improvements:

- Increased spatial resolution, from 5 to 2.5 km, using a regular latitude–longitude grid with 0.025º spacing.

- Enhanced cloud detection for GOES and HMWR sensors, accomplished through the integration of the latest version of the Nowcasting-SAF (NWC-SAF) GEO software package v2025.1.
- Comprehensive algorithm refactoring, ensuring improved efficiency and internal consistency.
- Full historical reprocessing covering the period 2018–2025.

An extensive quality assessment is carried out to evaluate the performance of the CLMS LST V3.0 hourly and 10-daily products over the study period from January 2018 to December 2025, depending on the availability of additional LST datasets used for the assessment. The analyses include direct comparisons with in-situ observations from the Copernicus Ground-Based Observations for Validation (GBOV) service and performance assessments using other independent global LST datasets, together with cross-comparisons against CLMS LST V_{PREV}. The validation against GBOV in-situ measurements provides a quantitative assessment of the CLMS LST product's accuracy and precision, while the comparison with other independent LST datasets allows an evaluation more focused on the temporal stability and overall compliance with the CLMS LST product quality requirements.

The results demonstrate a distinction in product consistency following the updates implemented in V3.0 for the reprocessing period of eight years. Enhanced scores are observed in the validation against 37 in-situ GBOV reference measurements, as well as improved performances when comparing the reprocessed dataset with ERA5 skin temperature (SKT) reanalysis and independent ESA Climate Change Initiative (CCI) LST estimates derived from the Sea and Land Surface Radiometer sensor onboard Sentinel-3A (SLSTR-A). These highlight more coherent temporal and spatial behaviours in V3.0 than V_{PREV}. The in-situ hourly validation (2018–2024) indicates that V3.0 produces overall lower bias (-0.43 K) and absolute residuals (1.76 K), while a comparable root-mean-squared difference (3.24 K) is obtained relative to V_{PREV}. Conformity with the product's requirements show that 53 % of V3.0's satellite/in-situ matchups meet the threshold requirement (i.e. absolute error below or equal to 2 K) while 29 % meet the goal requirement (i.e. absolute error below or equal to 1 K). The in-situ validation also includes the CLMS 10-daily products derived from the hourly datasets, namely the LST10 daily-cycle (LST10-DC) and synthesis (LST10-Syn) products. Results show that V3.0 performs consistently well in relation to V_{PREV}, with median 10-daily LST estimates exhibiting lower biases and comparable RMSD values.

Spatial consistency is analyzed by performing a cross-comparison of V3.0 with V_{PREV} for two separate periods (2018–2020 and 2021–2025). Results show that differences are substantially reduced after 2021. This is particularly observable over the MSGs domain, following the full integration of the IODC data in the CLMS LST product, indicating more stable and spatially consistent LST estimates with V3.0.

Temporal stability is assessed by comparing V3.0 against other independent products, namely ERA5 SKT (2018–2025) and SLSTR-A LSTs (2018–2024) using anomaly-based monthly mean

differences. Against ERA5, results indicate reduced global discrepancies and enhanced stability with V3.0, showing systematically lower anomaly deviations and weaker trends relative to V_{PREV}, reflecting enhanced temporal consistency. This becomes particularly evident after 2021, following the introduction of dynamic vegetation-based surface emissivity at 1 km and snow cover-based emissivity refinements, specially over the GOES-16 and HMWR components, where V_{PREV} exhibits significant temporal discontinuities. For the comparison against LST estimates from the SLSTR-A, the anomaly-based analysis highlights a gradual reduction of daytime differences (-0.45 K per decade) and small positive trend (+0.11 K per decade) for the nighttime period.

The spatial and temporal assessment of V3.0 was further carried out by performing a qualitative analysis using case studies of severe heatwave in Europe and southeastern South America. These illustrate that median LST10-Syn anomalies, combined with the Thermal Condition Index (TCI), effectively capture the spatial distribution and evolution of thermal stress, highlighting the added value of the 10-daily products for monitoring applications.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The document includes the validation of CLMS Land Surface Temperature (LST) V3.0 and examines the impact of the updates implemented in this new version compared to the previous operational release (V_{PREV}). The updates in V3.0 include:

- Increased spatial resolution, from 5 to 2.5 km, using a regular latitude–longitude grid with 0.025º spacing.
- Enhanced cloud detection for the Geostationary Operational Environmental Satellite (GOES) and Himawari (HMWR) sensors, accomplished through the integration of the latest version of the NWC-SAF GEO software package v2025.1.
- Comprehensive algorithm refactoring, ensuring improved efficiency and internal consistency.
- Full historical reprocessing covering the period from 2018–2025.

The overall assessment is based on LST retrievals covering the 8-year period from January 2018 to December 2025. It evaluates the quality of the new V3.0 in terms of accuracy and precision through comparisons with in-situ observations, as well as temporal and spatial consistency with comparisons against other global LST products. The analysis includes all satellite and in-situ datasets available upon the time of this report.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the requirements, and the expected product performance.
- Chapter 3 provides an overview of the CLMS LST hourly and 10-daily products, and details the changes introduced in version 3.0.
- Chapter 4 describes the validation methodology and includes the list of in-situ stations and statistical metrics used for the evaluation analysis.
- Chapter 5 presents the hourly and 10-daily evaluation results against in-situ data, as well as performance assessments by comparing V3.0 with other independent global LST products, organized in dedicated sections.
- Chapter 6 summarizes the main conclusions of the study and provides recommendations.
- Chapter 7 contains the appendix with additional supporting material.
- Chapter 8 lists references cited in the report.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_PQAD	Product Quality Assurance Document of the Copernicus Global Land Service
CGLOPS1_ATBD_FCOVER300m-V2	Algorithm Theoretical Basis Document of the FCOVER 300m Version 2.0 product
CGLOPS1_PUM_FCOVER1km-V2	Product User Manual of the FCOVER 1km Version 2 product
CGLOPS2_ATBD_SCE-NHEMI-1km_V1	Algorithm Theoretical Basis Document of the Northern Hemisphere Snow Cover Extent 1km V1.0 product
CGLOPS_QAR_LST-V2.0	Quality assessment report containing the results of the scientific validation of the Land Surface Temperature V2.0 product
CGLOPS_QAR_LST-V2.1	Quality assessment report containing the results of the scientific validation of the Land Surface Temperature V2.1 product
CGLOPS1_QAR_LST10-V2.1	Quality assessment report containing the results of the scientific validation of the 10-daily Land Surface Temperature Version 2.1 product

Available at: https://land.copernicus.eu/en/technical-library

1.3.3 Output

Document ID	Descriptor
CGLOPS1_ATBD_LST-V3.0	Algorithm Theoretical Basis Document for Land Surface Temperature Version 3.0 product
CGLOPS1_ATBD_LST10-V3.0	Algorithm Theoretical Basis Document of the 10-daily Land Surface Temperature Version 3.0 product
CGLOPS1_PUM_LST-V3.0	Product User Manual summarizing all information about the LST Version 3.0 product

Available at: https://land.copernicus.eu/en/technical-library

1.3.4 External documents

Document ID	Descriptor
GCOS#245	The 2022 GCOS ECVs Requirements

Available at: https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245?offset=1

GBOV-ATBD-RM2-RM8-RM9	Ground-Based Observations for Validation – Algorithm Theoretical Basis Document – Land Surface Temperature products

Available at: https://gbov.acri.fr/api/v1/document_versions/2/document

NWC-CDOP2-GEO-MFL-SCI-VR-Cloud_v1.0	Scientific and Validation report for the Cloud Product Processors of the NWC/GEO, Issue 1, Rev. 0, 2016

Available at: http://www.nwcsaf.org/AemetWebContents/ScientificDocumentation/Documentation/GEO/v2016/NWC-CDOP2-GEO-MFL-SCI-VR-Cloud_v1.0.pdf

JMA, 2017	Himawari-8/9 - Himawari Standard Data User's Guide, v1.3. Tokyo

Available at: https://www.data.jma.go.jp/mscweb/en/himawari89/space_segment/hsd_sample/HS_D_users_guide_en_v13.pdf

U.S. National Ice Center, 2008 *IMS Daily Northern Hemisphere Snow and Ice Analysis at 1 km, 4 km, and 24 km Resolutions*. (G02156, Version 1). Boulder, Colorado USA. National Snow and Ice Data Center. https://doi.org/10.7265/N52R3PMC

Available at: https://nsidc.org/data/g02156/versions/1

2 REQUIREMENTS

According to the applicable document [AD1], the LST product specific technical details and requirements are summarized below:

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

PRODUCT SPECIFICATION	
Geometric Properties:	
Geolocation precision	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetic datum	WGS84
Geographic projection	Regular latitude–longitude grid
Geographic coverage	Global

LST – LAND SURFACE TEMPERATURE	
The Land Surface Temperature (LST) is the radiative skin temperature of the land surface, as measured in the direction of the remote sensor [K].	
Spatial resolution	5 km
Time span:	Hourly and 10-day period
Timeliness:	Hourly: within 4 hours; within 2 days (optimally 1 day) after the end of 10-day period.
Uncertainty (2 sigma)	Threshold: 2 K, Goal: 1 K
Stability (per decade)	Threshold: 0.3 K, Goal: 0.1 K

2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data structure

Data coding[1] shall be compatible with the CLMS products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.

Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product.
- The per-pixel date of the individual measurement or the start-end dates of the period covered.
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the CLMS production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current CLMS component (operational product data formats and archive data formats) and other Copernicus services, all datasets shall be available in NetCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the Contracting Authority during the Framework Contract.

2.2.4 Uncertainties and validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Levels of uncertainty: the error estimates will account for the uncertainty of the algorithms themselves. Stability may relate to the existence of a trend between the product and reference ground measured or modelled data.

Validation of the products shall conform to at least the Committee on Earth Observation Satellite (CEOS) Land Product Validation (LPV) standards (https://lpvs.gsfc.nasa.gov/). Wherever appropriate, the bio-geophysical variables shall be validated and compared to CEOS Calibration/Validation (CAL/VAL) datasets and/or Ground-Based Observations for Validation (GBOV, https://land.copernicus.eu/en/products/GBOV) of CLMS when biophysical parameters are available.

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

To ensure continuity with the current operation, the same biomes as already used in the CLMS component shall be used.

2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu.

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts.
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the CLMS component have been based on SPOT VGT, PROBA-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary (GEO) and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary datasets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the CLMS component dissemination. The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

3 REVIEW OF CLMS LST AND CHANGES IN V3.0

3.1 ESTIMATION OF HOURLY LST

The CLMS LST product is derived from a constellation of GEO satellites, operated by different international agencies (Freitas et al. 2013). Such operational missions have evolved since the original design of the CLMS LST product, resulting in several adjustments made over time, justifying the release of a new product version. The original product relied on data from Meteosat Second Generation (MSG), Geostationary Operational Environmental Satellite (GOES) and the Multi-Function Transport Satellite (MTSAT). Since then, MTSAT was replaced by the third-generation HMWR-8 platform in November 2015, while GOES-16 became operational in December 2017, taking over the GOES-East coverage. Although the MSG nominal 0º mission has remained unchanged, older MSG platforms have been reassigned to ensure the Indian Ocean Data Coverage (IODC) mission, which is currently maintained by MSG-2 positioned at 45.5° E longitude. This relatively recent demonstration mission allows to fill a previous data gap over the Middle East that existed in earlier product versions due to the lack of a suitable GEO instrument for that region when the CLMS LST product was initially developed. Data from the MSG 0° and IODC missions are provided by the Land Surface Analysis Satellite Application Facility (LSA-SAF), while GOES-16 and HMWR mission data are processed specifically for the CLMS LST product.

Both GOES-16 and HMWR-8 enable the use of a more accurate LST retrieval algorithm compared to their predecessors, since they provide the spectral channels required for the Generalized Split Windows (GSW) approach [CGLOPS1_ATBD_LST-V3.0]. This algorithm relies on a multiple linear regression to estimate LST from brightness temperatures of two adjacent thermal infraRed (TIR) channels and their corresponding surface emissivities. These regressions are calibrated for different classes of Total Column Water Vapour and Satellite Viewing Angle to include an implicit dependency on the optical path. A Dual Algorithm had to be used for the previous generation sensors, with a simpler formulation using only one TIR channel during daytime, while a combination of the TIR and Middle InfraRed (MIR) channels was used during nighttime to improve retrieval accuracy. Both missions have since been succeeded by newer platforms: GOES-19, launched in June 2024 and operational since April 2025, which retains the same core Imager; and HMWR-9, launched in November 2016 and operational since December 2022, which replaces HMWR-8 to correct and enhance the optical imager's performance – particularly across the visible bands (1–4) – therefore improving the overall accuracy of the data.

Since the inclusion of HMWR-8 in CLMS product V1.2, further research has focused on optimizing the calibration of GSW algorithms, such as the one used in CLMS LST (Martins et al., 2016). This improved methodology was adopted in V2.0, where both the GOES-16 and HMWR retrievals rely on a more comprehensive calibration database that accounts for a wider range of land surface types, atmospheric and viewing geometry conditions. In contrast, LST retrievals from LSA-SAF for the

MSG missions (0º and IODC) follow a simpler, yet comprehensive, calibration approach for their operational products (Trigo et al., 2008a).

Surface emissivity is a key input for all LST retrievals and it is estimated using the Vegetation Cover Method. In this approach, for each land cover type, bare ground and vegetation emissivities are derived from a comprehensive spectral emissivity library (Trigo et al., 2008b). The channel emissivity for each pixel is then obtained by weighting these two components according to the fraction of vegetation cover (FCOVER). While the LSA-SAF LST retrieval uses daily FCOVER estimates from LSA-SAF (García-Haro et al., 2005; Roujean et al., 1992; Trigo et al., 2008b), the GOES-16 and HMWR missions use the CLMS FCOVER product, which is available at a dekadal frequency. In version V2.0, a seasonal climatology of CLMS FCOVER 1 km product was applied [CGLOPS1_PUM_FCOVER1km-V2], while version V2.1 introduced the near-real time (NRT) FCOVER product 300 m [CGLOPS1_ATBD_FCOVER300m-V2] derived from Sentinel-3 observations.

In version V2.1, snow information is introduced to correct channel emissivities in areas where non-permanent snow is detected. Since the retrieval algorithm uses a set of emissivities for both bare ground and vegetation according to land cover class, any pixel identified as snow-covered is simply reclassified as snow/ice. The CLMS Snow Cover Extent (SCE) 1 km product [CGLOPS2_ATBD_SCE-NHEMI-1km_V1] is reprojected to the GEO projection used for the GOES-16 and HMWR LST retrievals. All pixels where the average SCE value is greater than zero are corrected accordingly. The predefined emissivity values for snow/ice are further adjusted as function of the satellite viewing angle. The final pixel emissivity is then estimated as a weighted average between the snow emissivity and the emissivity corresponding to the original land cover type.

Due to the revised swath configurations of the sensors used in the CLMS LST product, the definition of hourly timeslots was updated. The use of GOES-16 data with the previous configuration resulted in acquisition time differences higher than 20 minutes within the same hourly file. Such temporal inconsistency was minimized by redefining the timeslot configuration. The change is particularly relevant in regions where multiple sensors contribute to the final LST estimate, since synchronized acquisition times allow to reduce discontinuities across overlapping areas. Consequently, some temporal discontinuities can be observed when combining time-series from product V1.2, V2.0, V2.1, and V2.2, especially for regions over the MSG disk, where the timeslot reference changed from T-15 minutes to T (T being the reference time for each file; CGLOPS_QAR_LST-V2.1).

3.2 CHANGES IN V3.0

With the introduction of version 3.0, several major upgrades were implemented following the previous operational version, including an increase in spatial resolution, enhanced input components (cloud masking and snow cover handling), and a complete refactoring of the LST retrieval algorithm

[CGLOPS1_ATBD_LST-V3.0]. These enhancements were applied consistently over the full reprocessing period (January 2018–December 2025) and are to be maintained for the NRT production moving forward.

Previous versions of the CLMS LST product adopted a 5 km spatial resolution, primarily to harmonize data from the various GEO platforms available at the time. Legacy sensors, such as MTSAT and GOES Imager, provided TIR observations at approximately 4 km (at nadir), while MSG offered a slightly finer resolution of about 3 km. To ensure cross-platform consistency, the global product was therefore generated on a coarser 5 km grid. With the introduction of the recent generation of sensors, this constraint was removed. Since 2018, both GOES (ABI) and HMWR (AHI) instruments have delivered TIR data at 2 km resolution (at nadir), while MSG continued to operate at 3 km. Taking advantage of these advances, CLMS LST V3.0 now delivers global LST estimates at 2.5 km on a regular latitude–longitude grid. This optimization improves the matching between native sensor resolutions and reduces the loss of spatial detail, therefore increasing the product's value for downstream applications.

In addition to incorporating the latest updates from the previous operational version (addressing the discontinuities identified between V1.2 to V2.2), V3.0 introduces several fundamental enhancements. These include an upgraded cloud detection for GOES and HMWR sensors based on the most recent version of the NWC-SAF GEO software package v2025.1 [NWC-CDOP2-GEO-MFL-SCI-VR-Cloud_v1.0]. Compared to the previous implementation, which relied on threshold-based tests using a limited set of spectral channels, the updated scheme exploits the higher spectral resolution of the new generation sensors ABI and AHI (respectively). Moreover, the NWC-SAF software applies more advanced tests, including improved detection of low-level and thin high-clouds, as well as spatial and temporal consistency checks. Although this approach leads to more reliable cloud screening, some limitations can exist since cloud detection is not perfect and residual cloud contamination (i.e. missed clouds) may still be present in LST retrievals. These enhancements also include a refined snow cover treatment that combines the CLMS SCE 1 km product [CGLOPS2_ATBD_SCE-NHEMI-1km_V1] with the Interactive Multisensor Snow and Ice Mapping System (IMS) product [U.S. National Ice Center, 2008] for gap-filling over the Northern Hemisphere, and further refactoring of the LST retrieval algorithm.

Another change concerns the handling of data availability. In V_{PREV}, more restrictive conditions were implemented during the LST processing, which reduced the number of valid retrievals. In V3.0, such constraints are removed, resulting in more observations to be retained. Therefore, any additional filtering based on fraction of available data is left for the user to decide. More detailed information on CLMS LST V3.0 can be found in the product's PUM [CGLOPS1_PUM_LST-V3.0] and ATDB [CGLOPS1_ATBD_LST-V3.0].

3.3 LST10-DAILY CYCLE

The LST10 – Daily Cycle (LST10-DC) is a 10-day composite of LST product for hourly timeslots (00 to 23UTC) where the following parameters are provided:

- Maximum, median and minimum of valid LST values observed during the compositing period per timeslot.
- Fraction of valid observations per timeslot used to calculate the statistical parameters.

3.4 LST10-SYNTHESIS

The synthesis of 10-day LST consists of:

- Maximum, median and minimum LST values observed during the compositing period (regardless of the observation time).
- Thermal Condition Index (TCI).
- Fraction of valid LST values (used to calculate statistics) in the compositing period. In this context, a valid LST value corresponds to a clear-sky retrieval after outlier removal.

Each value of the LST10-Synthesis is assumed to be representative of the LST over the dekad regardless of the observation time. A detailed description of the reference climatology is presented in the [CGLOPS1_ATBD_LST10-V3.0].

4 VALIDATION METHODOLOGY

The validation methodology is based on a direct comparison between hourly and 10-daily satellite-derived LST estimates and in-situ observations from the GBOV service. The analysis covers the period from January 2018 to December 2024. Comparisons between in-situ and CLMS LST estimates are performed at the satellite pixel closest to each station location, and only matchups with an absolute difference in acquisition time less than 3 minutes (Trigo et al., 2008a) are considered valid, particularly for stations measuring at 1 or 5-minute frequency. For stations reporting at 30-minute intervals (e.g. in Australia), the nearest satellite timeslot is used. To ensure consistency between V3.0 and V_{PREV}, only timeslots with identical data availability in both versions are retained for the matchup analysis. According to the product requirements (section 2), the resulting satellite/in-situ LST matchup time-series are evaluated at each station using a set of statistical metrics designed to quantify accuracy, precision, and associated error of the CLMS LST estimates, as well as conformity tests performed at the pixel level, as described in the following sections 4.2 and 4.3, respectively.

Evaluation analyses are also carried out to assess the temporal stability of V3.0 using independent datasets (section 4.4). These are complemented with qualitative assessments towards the spatial consistency of the CLMS product. In particular, seasonal LST distributions from V3.0 and V_{PREV} are compared over two separate periods (2018–2020 and 2021–2025). This approach reflects differences in V_{PREV} input configuration, since important updates – such as the introduction of dynamic vegetation-based surface emissivity and the integration of IODC data – were only implemented 2021 onwards, while V3.0 includes these components consistently over the full period. In addition, V3.0 LST anomalies for two heatwave case studies are also presented.

The corresponding data availability for V3.0, V_{PREV}, and the individual GEO components is provided in Figure A1 (Appendix).

4.1 IN-SITU STATIONS

The validation of CLMS LST V3.0 is based on the most recent GBOV reference measurements for LST datasets (version 2.2, released in September 2023). The GBOV service collects raw measurements from each ground station in whatever format the source data presents itself, and provides harmonized LST reference values derived according to the instrumentation available at each site. GBOV output files are distributed in a standardized comma-separated values (CSV) format, facilitating their integration and direct comparison with satellite-derived estimates. The methodologies used to derive in-situ LSTs for each station are described in detail in the GBOV product ATBD (GBOV-ATBD-RM2-RM8-RM9). It is worth noting that by the time of this analysis, no in-situ data was available from the GBOV portal for 2025, therefore the direct validation is limited to the period up to December 2024.

Given the high spatial and temporal variability of LST, direct validation against in-situ measurements is challenging. Therefore, an initial data curation is performed to select only robust GBOV sites suitable for satellite LST validation. This pre-selection includes a comprehensive quality-control and data availability checks, from which 37 stations are retained from an initial total of 55 (see Table 1). Specifically, this selection follows four main criteria: (i) at least one full year of valid data within 2018–2024; (ii) location within the field of view of the corresponding GEO platform; (iii) no instrumentation issues affecting local measurements; (iv) consistency with surface representativeness at the satellite pixel scale. The latter two criteria are assessed through a detailed inspection of raw in-situ time-series and examination of the station surrounding environment using high-resolution orography, land cover maps, and Landsat-based LST retrievals, implemented via the Google Earth Engine algorithm developed by Ermida et al. (2020).

These selected stations are geographically distributed over the United States, Europe, Africa, Asia, and Australia, covering a wide range of land cover types, vegetation conditions, elevations and climate zones (Figure 1). Such diversity ensures a robust assessment of the CLMS LST V3.0 dataset under varying surface and atmospheric conditions. More details on the characteristics of each GBOV station are provided in Table 1.

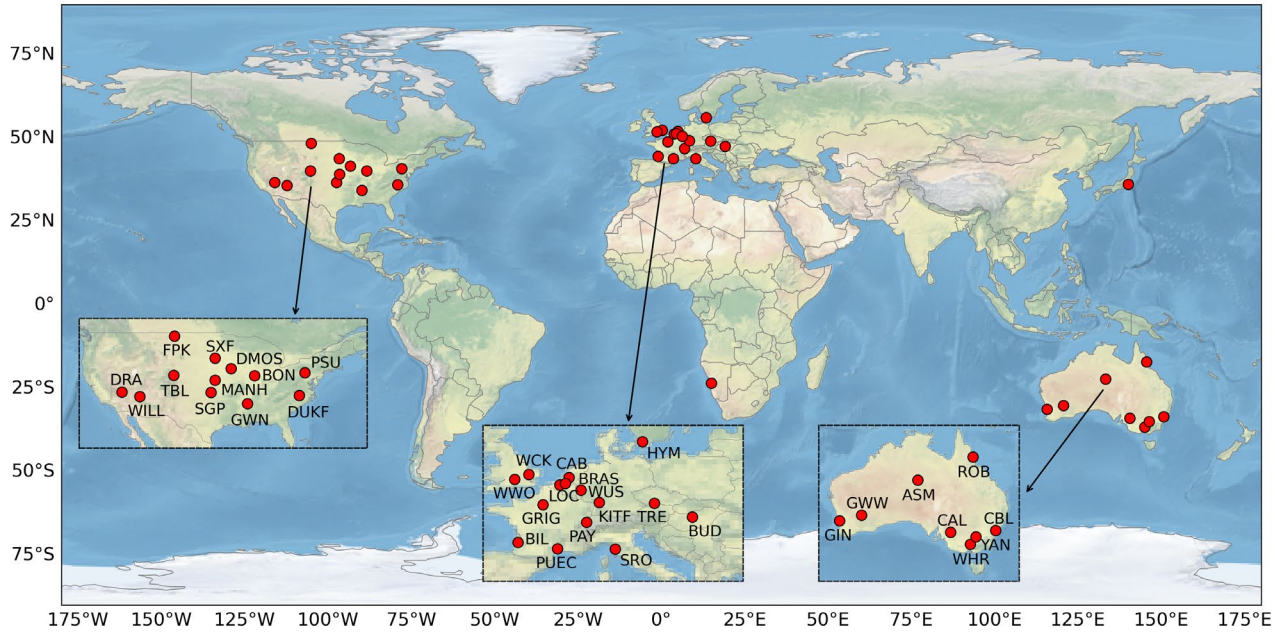


Figure 1 - Geolocation of all LST reference measurements from the GBOV service that are used for the validation of CLMS LST V3.0.

Table 1 - List of GBOV stations used in this report. Land cover information was retrieved from http://maps.elie.ucl.ac.be/CCI/viewer/index.php, and Köppen-Geiger climate classification data was obtained from http://koeppen-geiger.vu-wien.ac.at/.

Name	Code	Latitude	Longitude	Elev. (m)	CCI Land Cover	Köppen climate classification
Fort Peck (USA)	FPK	48.31° N	105.10° W	634	130 (Grassland)	Arid, steppe, cold (BSk)
Des Moines (USA)	DMOS	41.56° N	93.29° W	281	30 (Cropland)	Snow, humid, hot summer (Dfa)
Bondville (USA)	BON	40.05° N	88.37° W	230	11 (Cropland)	Temperate, humid, hot summer (Cfa)
Manhattan (USA)	MANH	39.10° N	96.61° W	347	130 (Grassland)	Temperate, humid, hot summer (Cfa)
Williams (USA)	WILL	35.76° N	112.34° W	1826	120 (Shrubland)	Arid, steppe, cold (BSk)
Southern Great Plains (USA)	SGP	36.61° N	97.49° W	318	11 (Cropland)	Temperate, humid, hot summer (Cfa)
Duke Forest (USA)	DUKF	35.97º N	79.09º W	170	90 (Tree cover)	Temperate, humid, hot summer (Cfa)
Desert Rock (USA)	DRA	36.62° N	116.02° W	1006	120 (Shrubland)	Cold arid (BWk)
Rock Springs (USA)	PSU	40.72º N	77.93° W	376	11 (Cropland)	Temperate, humid, hot summer (Cfa)
Sioux Falls (USA)	SXF	43.73° N	96.62° W	473	11 (Cropland)	Cold, humid (Dfa), hot summer (Cfa)
Table Mountain (USA)	TBL	40.12° N	105.24° W	1689	11 (Herbaceous cover)	Arid, steppe, cold (BSk)
Goodwin Creek (USA)	GWN	34.25° N	89.87° W	98	40 (Mixed Vegetation)	Temperate, humid, hot summer (Cfa)
Bilos (France)	BIL	44.49º N	0.96 º W	38	70 (Tree cover)	Marine west coast (cfb)

Name	Code	Latitude	Longitude	Elev. (m)	CCI Land Cover	Köppen climate classification
Payerne (Switzerland)	PAY	46.81° N	6.94° E	637	10 (Cropland)	Temperate, humid (Cfb)
Cabauw (The Netherlands)	CAB	51.97° N	4.93° E	0	130 (Grassland)	Temperate, humid (Cfb)
Puechabon (France)	PUEC	43.74° N	3.60° E	261	70 (Tree cover)	Mediterranean, hot dry summer (Csa)
SanRossore2 (Italy)	SRO	43.73° N	10.29° E	12	70 (Tree cover)	Meditteranean, hot summer (Csa)
Budapest (Hungary)	BUD	47.43 N	19.18° E	137	190 (Urban area)	Temperate, humid (Cfb)
Grignon (France)	GRIG	48.84° N	1.95° E	137	11 (Herbaceous cover)	Temperate, warm summer (Cfb)
Trebon (Czech Republic)	TRE	49.03° N	14.77° E	426	130 (Grassland)	Humid continental (Dfb)
Lochristi (Belgium)	LOC	51.11° N	3.85° E	6	11 (Herbacious cover)	Marine weat coast (Cfb)
Brasschaat (Belgium)	BRAS	51.31° N	4.52° E	16	110 (Herbacious cover)	Temperate, warm summer (Cfb)
Wytham Woods (UK)	WWO	51.78° N	1.34° W	125	60 (Tree cover)	Temperate, humid (Cfb)
Hyltemossa (Sweden)	HYM	56.10° N	13.42° E	114	70 (Tree cover)	Continental, humid, warm summer (Dfb)
WickenFen (UK)	WCK	52.31° N	0.28° E	4	11 (Herbacious cover)	Temperate, warm summer (Cfb)
Wustebach (Germany)	WUS	50.50° N	6.33° E	624	11 (Tree cover)	Marine west coast (Cfb)
KIT Forest (Germany)	KIT	49.09° N	8.42° E	118	11 (Tree cover)	Temperate, warm summer (Cfb)
Gingin (Australia)	GIN	31.38° S	115.71° E	50	62 (Tree cover)	Warm temperate, hot dry summer (Csa)
Great Western Woodlands (Australia)	GWW	30.19° S	120.65° E	449	122 (Shrubland deciduous)	Arid steppe (BSh)
Alice Springs Mulga (Australia)	ASM	22.28° S	133.25° E	607	120 (Shrubland)	Desert, cold, arid (BWk)
Robson Creek (Australia)	ROB	17.12° S	145.63° E	680	50 (Tree cover)	Temperate, humid, hot summer (Cfa)
Calperum (Australia)	CAL	34.00° S	140.59° E	64	120 (Shrubland)	Arid, steppe, cold (BSk)
Whroo (Australia)	WHR	36.67° S	145.03° E	144	50 (Tree cover)	Temperate, humid, hot summer (Cfa)
Yanco (Australia)	YAN	34.99° S	146.29° E	119	10 (Cropland)	Arid, steppe, cold (BSk)
Cumberland Plain (Australia)	CBL	33.62° S	150.72° E	24	120 (Shrubland)	Temperate, humid, hot summer (Cfa)
Gobabeb (Namibia)	GOB	23.56° S	15.04° E	406	200 (Bare)	Arid, desert (BWh)
Tateno (Japan)	TAT	36.06° N	140.13° E	26	10 (Cropland)	Warm, humid, warm summer (Cfa)

4.2 STATISTICAL METRICS

The evaluation of satellite-derived LST estimates is carried out using a set of robust statistical metrics computed from the obtained satellite/in-situ matchup pairs, which are spatially and temporally synchronized. This includes the median error, i.e. the bias (μ), as means to quantify the accuracy:

$$\mu = median(LST_{sat} - LST_{obs}), \quad (1)$$

where LST_{sat} denotes the satellite-derived LST estimate, and LST_{obs} the corresponding in-situ observation. For the precision, the median of the absolute residuals (σ) is used:

$$\sigma = median(|(LST_{sat} - LST_{obs}) - \mu|). \quad (2)$$

To account for both magnitude and distribution of the errors, the root-mean-squared-difference (RMSD) is also applied:

$$RMSD = \sqrt{\frac{\sum(LST_{sat} - LST_{obs})^2}{N}}, \quad (3)$$

where N is the number of valid matchup samples. To quantify the uncertainty associated with sampling in the metrics calculations, 95 % confidence intervals are computed for each statistical metric. These intervals are obtained through a bootstrap resampling approach (with 1000 iterations) applied to the series of matchup data. This method provides a robust estimation of uncertainty while accounting for the inherent variability in the observations, having been used successfully in previous studies (e.g. Dutra et al., 2014).

4.3 CONFORMITY TESTS

To further assess the performance of the CLMS LST product, conformity tests are performed by considering the apparent error at each pixel *i* (ε_i), following the product specifications (section 2). This error is defined as:

$$\varepsilon_i = LST_{sat} - LST_{insitu}. \quad (4)$$

With Δ representing the maximum permissible error, the conformity status is determined according to the following criteria:

- Conform: when the absolute error is less than or equal to the maximum permissible error:

$$|\varepsilon_i| \leq \Delta\ ; \quad (5)$$

- Non-Conform: when the absolute error exceeds the maximum permissible error:

$$|\varepsilon_i| > \Delta\ ; \quad (6)$$

The conformity results are expressed as the percentage of conforming and non-conforming cases for each station, providing a complementary measure of compliance with the product accuracy requirements.

4.4 EVALUATION USING INDEPENDENT DATASETS

4.4.1 ERA5 reanalysis

The temporal stability of the hourly CLMS LST product is assessed using the skin temperature (SKT) parameter from ERA5 reanalysis (Hersbach et al., 2020) as an independent dataset over the period January 2018 to December 2025. ERA5 SKT datasets are produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) and are obtained from the Copernicus Climate Change Service (C3S) Climate Data Store (https://cds.climate.copernicus.eu/). This surface parameter has been shown to provide a suitable large-scale proxy for satellite-derived LST consistency assessments (Trigo et al., 2015). To ensure consistent comparisons between datasets, both CLMS LST V3.0 and V_{PREV} are spatially aggregated to match ERA5's regular 0.25º latitude–longitude grid. For each grid cell, hourly LST values are used to derive the monthly mean diurnal cycle, from which the minimum, mean, and maximum LST values are extracted, provided that at least 30 % of the hourly observations are valid. These quantities are then averaged to obtain monthly mean values.

To better characterize potential differences between V3.0 and V_{PREV}, anomaly differences of the monthly means are analysed by removing the mean seasonal cycle from both time-series. Temporal trends over the study period are then estimated using a non-parametric Mann-Kendall test, implemented through the *pymannkendall* Python package, with the associated slope used to quantify the magnitude of the trends. The analysis is performed at both global and regional scales, allowing a more detailed assessment of the CLMS LST performance within each GEO domain (i.e. GOES-16, MSG 0º + IODC, and HMWR).

4.4.2 Sentinel-3A SLSTR LST

To further assess the consistency of CLMS LST V3.0, we compare it against the clear-sky SLSTR LST product from Sentinel-3A (SLSTR-A), part of the ESA CCI LST dataset (https://climate.esa.int/en/projects/land-surface-temperature/). Specifically, Level-3 supercollated (L3S) global datasets version 4.00 (Ghent et al., 2025) are extracted from the Joint Analysis System Meeting Infrastructure Needs (JASMIN) storage facility (https://jasmin.ac.uk/) on a native 0.01º regular latitude–longitude grid as daily daytime and nighttime NetCDF files, being then aggregated to the 0.025º grid of the CLMS LST product. It is worth noting that only Sentinel-3A is considered to ensure consistency and data availability, since the approximately 30-minute overpass difference relative to Sentinel-3B would provide small added value at the temporal scale of this analysis. For the matchup analysis, CLMS LST V3.0 daytime and nighttime values are temporally interpolated between consecutive hourly fields to match the exact Sentinel-3A acquisition times. Only spatially collocated pixels with at least 30 % valid observations in both datasets are retained. Similarly to the

ERA5-based assessment, this analysis uses anomaly differences of the monthly means between V3.0 and SLSTR-A to evaluate temporal stability. The comparison covers the period from January 2018 to February 2024; for 2024, only January and February are included due to limited SLSTR-A L3S data availability from the JASMIN storage facility. This approach allows a robust assessment of the temporal stability of V3.0 and highlights its added value relative to polar-orbiting LST retrievals.

5 RESULTS

5.1 IN-SITU COMPARISONS

5.1.1 Time-series analysis

An initial consistency check is performed by visually inspecting hourly time-series to compare the variability of both in-situ values and their corresponding satellite matchups in time. This analysis includes a side-by-side comparison between both V3.0 and V_{PREV} datasets. An example is provided Figure 2 where some consecutive days with snow conditions occur in the first dekad of January 2023, specifically for station FPK (USA). The use of a snow mask to derive emissivity was only introduced in CLMS LST V2.1 on December 2022. However, LST estimates that had been generated since then relied only on the CLMS SCE product, which usually has large areas of missing data, mainly due to cloud-contaminated pixels. In such areas, the emissivity falls back to snow free conditions. In contrast, the latest CLMS LST V3.0 combines the CLMS SCE with the IMS 4 km snow map to perform gap-filling, providing to more consistent LST estimates. According to the respective emissivity inputs (not shown here), this occurs on days 2, 4, 5 and 7, where missing snow data leads to increased emissivity fields in V_{PREV}. The impact of the different snow cover inputs is visible in Figure 2 where V_{PREV} shows higher deviations towards in-situ under such conditions in comparison to V3.0. Consequently, the V3.0 biases (Figure 2, bottom panel) are lower (±2 K) than the V_{PREV} biases (-2 to +4 K). During the analyzed 10-day period, there is a relatively low number of matchups between observation and satellite due to cloud coverage. In the last days of the dekad, comparable larger negative errors are observed for both versions, which are related to the combination of cloudy and snow conditions not identified by the cloud mask.

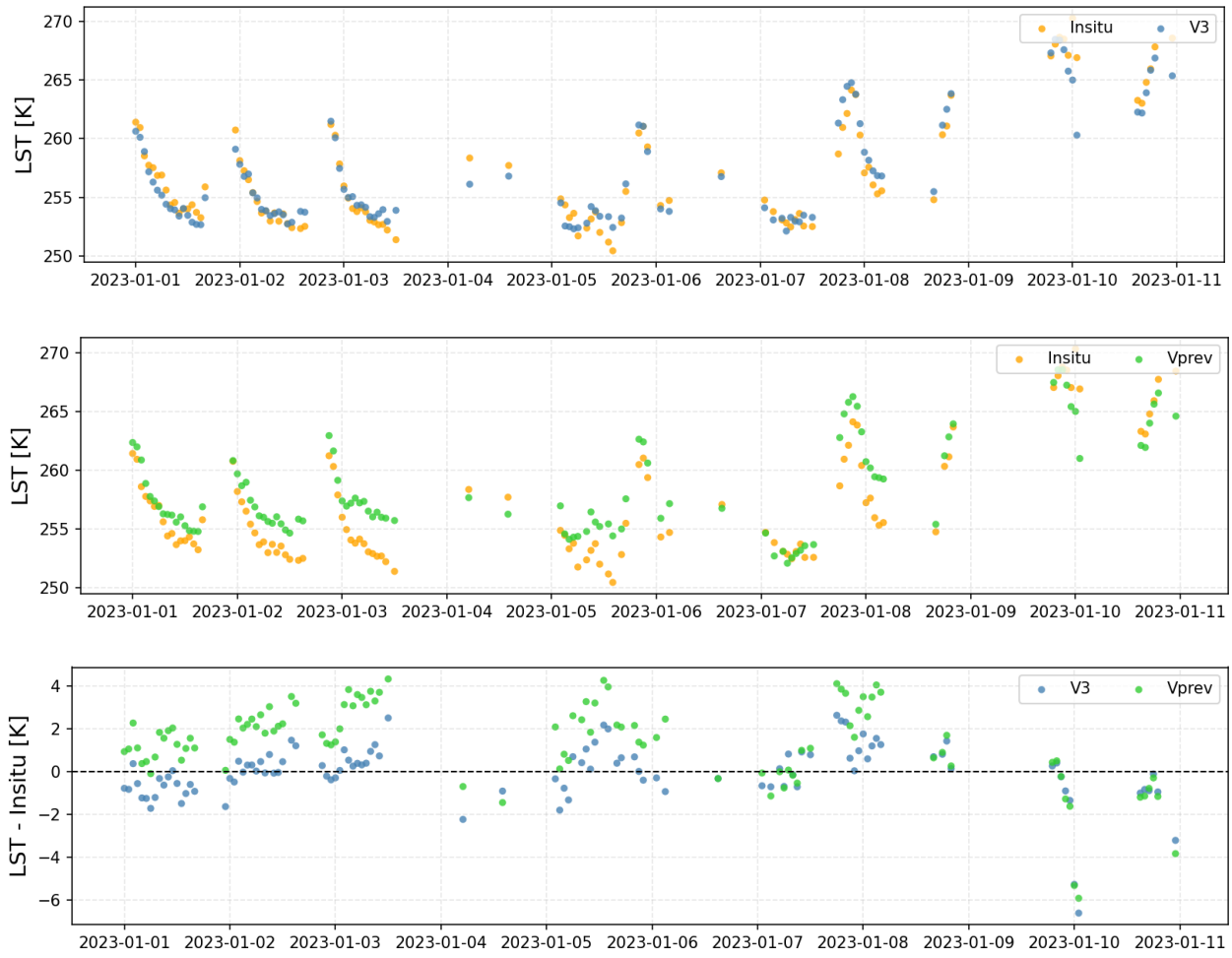


Figure 2 - Comparison between CLMS LST (in Kelvin) V3.0 (top) and V_{PREV} (middle) against in-situ observations for Fort Peck, USA (GBOV_FPK), during the first dekad of January 2023. The bottom panel presents the corresponding satellite/in-situ differences. Only valid matchup instances are shown. Each in-situ value is an average of the valid data in a ±3 minutes interval from the CLMS LST acquisition time.

In Figure 3, another comparison example is provided but for the first dekad of January 2024 in station GIN (Australia). Overall, very similar behaviours are observed during this period which are characterized by the presence of clear-sky conditions, except for second day due to the presence of clouds (as shown by the lack of satellite/in-situ matchups). When compared, both CLMS LST versions generally overestimate in-situ observations at mid-day while underestimations occur during the late afternoon-nighttime period. It is worth noting that during the latter period V3.0 shows lower deviations towards observations in comparison to V_{PREV}. A difference of about 0.5 K between both versions during this period is observable, being related to the use of inconsistent emissivity coefficient in V_{PREV} following the transition from HMWR-8 to 9, since these remained correct in V3.0. Overall,

these results show an improved representation of LST under different surface conditions with V3.0, which comprises refined surface emissivity inputs, allowing a more detailed characterization of surface emissivity and therefore accurate LST estimates compared to V$_{PREV}$.

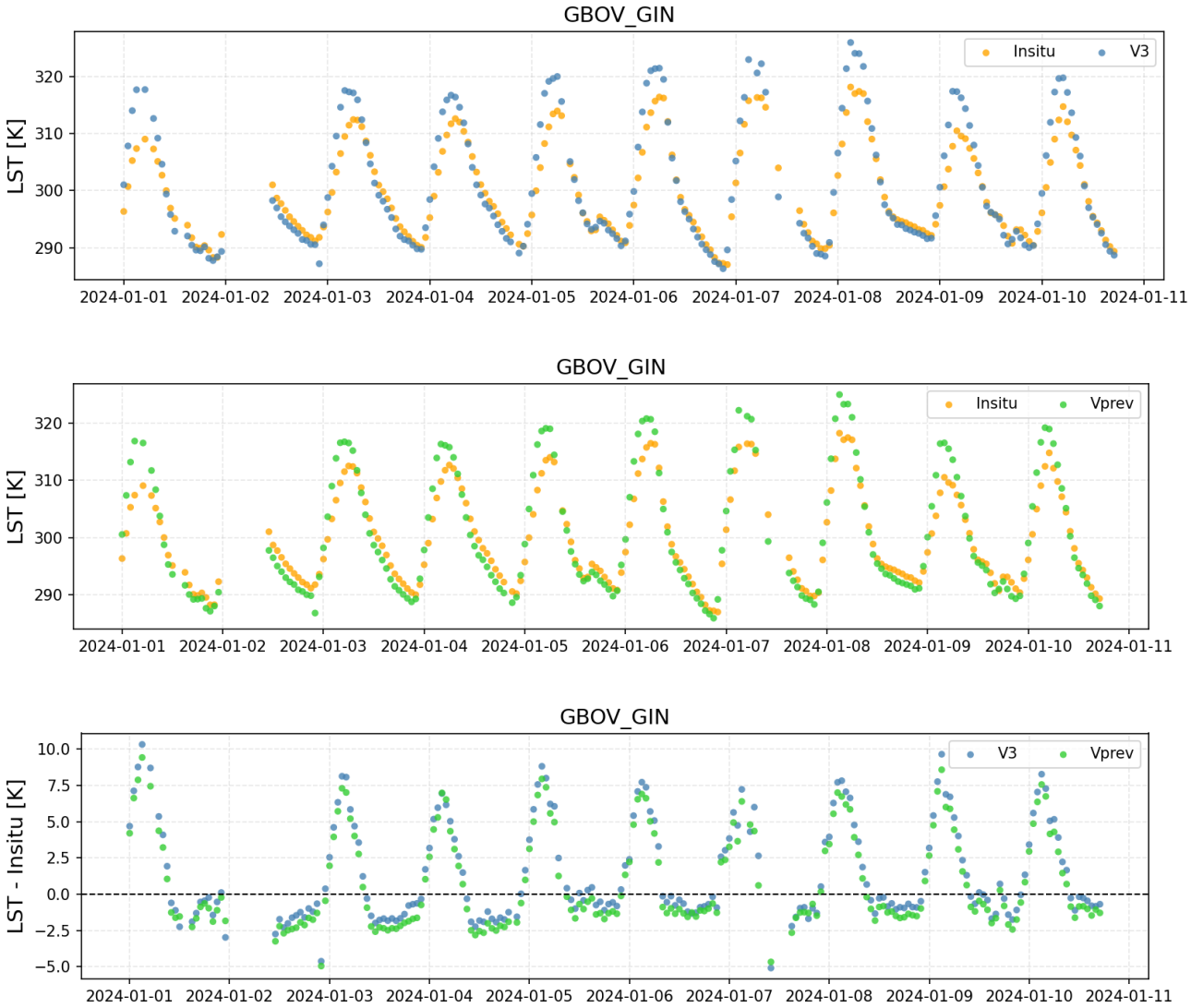


Figure 3 - Same as Figure 2 but for Gingin, Australia (GBOV_GIN), during the first dekad of January 2024.

5.1.2 Summary statistics

In Table 2, the statistical summary considering all available satellite/in-situ hourly matchups (independently for each GBOV station) is shown for the period January 2018–December 2024 using the performance metrics described in section 4 for the validation of the selected CLMS LST dataset.

In addition to these, conformity tests are included to inspect the percentage of cases that conform to the threshold specifications (i.e. when $|\varepsilon_i| \leq 2$ K) and goal specifications (i.e. when $|\varepsilon_i| \leq 1$ K), as described in section 2.

From the list of 37 GBOV stations, similar metrics can be found between V3.0 and V_{PREV} amongst the majority of the sites. Considering the total number of matchups from all stations (total row), the main differences between V3.0 and V_{PREV} can be highlighted over the product's accuracy, where the former shows a smaller negative bias of -0.43 K in comparison to the latter (-0.59 K). In terms of total precision and RMSD, the values obtained for V3.0 are 1.76 and 3.24 K, while for V_{PREV} are 1.81 and 3.23 K, respectively. Conformity tests show an overall conform percentage between satellite and observation matchups of 53 and 29 % in respect to threshold and goal requirements for V3.0, while for V_{PREV} 51 and 27 % are in conform with the corresponding requirements. The independent validation of V3.0 and $V_{PREV,}$ i.e. with different temporal matchups (total* row), shows similar results with lower accuracy and higher RMSD. This is expected due to the higher number of samples which are likely to be affected by cloud contamination.

Despite the similar behaviour between both versions towards in-situ, it is worth noting that there are also some stations where larger error discrepancies exist. For instance, in station PSU, there is an overall statistical improvement in V3.0, with particularly lower RMSD (1.98 K) than the one found for V_{PREV} (2.78 K). Since this station is located in a cropland field, the surrounding crops will affect local emissivity estimation. Therefore, for a correct quantification of LST it is essential to produce accurate emissivity estimates. A similar behaviour is observed in station FPK (surrounded by grassland), which is also known to experience frequent snow events during the winter periods. Other examples showing better statistics (accuracy, precision and uncertainty) with V3.0 are stations SGP, GWN, PUEC, WWO, WCK, ROB, CBL, and GOB. Other stations, like GWW and CAL, are characterized by significant lower biases with V3.0 but show marginal higher RMSD.

Larger errors can also be found in both CLMS LST versions. For instance, station DRA shows a very high negative bias (about -3 K) and a large RMSD (about 4 K) for V3.0 and V_{PREV}. This station has been associated to higher errors when compared to satellite LST estimates. For instance, Ghent et al. (2024) found a similar bias (-3.06 K), stating that this behaviour was likely related to the misrepresentation of surface heterogeneity on the selected satellite pixel in comparison to the in-situ LST. These representativeness issues have also been attributed to other stations such as stations DUKF, LOC, and CAL, in which, despite demonstrating reasonable biases (-0.37 to +1.65 K), these show larger RMSDs (4 to 4.5 K) in comparison to the remaining stations. Particularly, larger errors in stations DUKF and CAL have been observed to occur in previous quality assessment reports, where external factors such as vegetation cover variability are likely hindering hourly matchups between satellite and observation LST estimates. For the case of station LOC, this site is located in a highly heterogeneous area of crop fields close to urban areas, creating large discrepancies between the local measurements and the satellite's 2.5-km pixel scale.

Table 2 – Statistical summary for the 37 GBOV stations included in this study (2018–2024). Performance metrics are reported for both CLMS LST V3.0 and the previous operational version (V_{PREV}), based on same number of hourly matchups within the considered period. For each station, the statistical metrics (bias: μ, absolute residuals: σ, root-mean-squared difference: RMSD) and corresponding number of matchups are presented. Conformity with the LST product specifications is evaluated according to the defined threshold and goal criteria (see section 2). Global summary statistics computed using all hourly LST matchups from all stations are provided in the last two rows considering the same matchups for V_{PREV} and V3.0 (total row) and independent matchups (total* row), including the 95% confidence interval between brackets. LST values are expressed in Kelvin, while percentages are used for threshold and goal requirements. Best results for each criterion are highlighted in green.

Station code	μ [K]		σ [K]		RMSD [K]		Threshold [%]		Goal [%]		#matchups
	V3.0	V_{PREV}	V3.0	V_{PREV}	V3.0	V_{PREV}	V3.0	V_{PREV}	V3.0	V_{PREV}	V3.0 & V_{PREV}
FPK	-1.09	-1.15	1.17	1.26	2.27	2.39	66.50	62.30	37.34	33.68	21843
DMOS	-0.17	0.03	1.11	1.11	2.13	2.18	72.99	72.87	45.64	45.99	22786
BON	-0.58	-0.60	1.24	1.22	2.68	2.64	65.62	65.85	39.62	39.31	20194
MANH	0.62	0.29	1.24	1.26	2.32	2.30	66.47	68.74	37.15	40.07	26941
WILL	0.37	-0.08	1.31	1.37	2.58	2.55	66.92	66.16	39.07	37.96	33477
SGP	0.12	-0.16	1.15	1.17	2.31	2.36	72.58	72.39	43.36	44.76	21205
DUKF	1.65	1.52	3.15	3.02	4.51	4.35	23.51	24.44	11.02	11.23	24984
DRA	-3.06	-3.03	1.37	1.60	4.16	4.29	27.69	31.01	10.92	14.21	35473
PSU	0.07	0.17	1.27	1.91	1.98	2.78	71.51	51.87	40.42	27.95	18665
SXF	-0.92	-0.89	1.21	1.20	2.32	2.29	65.73	66.04	36.39	37.72	22333
TBL	-1.88	-1.97	1.34	1.34	3.04	3.07	50.18	48.36	25.90	24.36	28106
GWN	-0.29	-0.38	1.56	1.62	2.42	2.49	60.13	58.86	33.54	32.09	24151
BIL	-2.13	-2.08	1.69	1.69	3.76	3.72	30.13	31.06	12.24	12.53	17193
PAY	-0.67	-0.25	1.69	1.68	3.31	3.26	59.73	60.34	29.68	29.93	19241
CAB	-0.39	-0.34	1.08	1.10	2.65	2.63	72.91	72.16	46.90	46.59	18080
PUEC	1.29	1.30	1.26	1.27	2.81	2.82	58.02	57.80	33.13	33.08	12132
SRO	-1.43	-1.15	2.13	2.02	3.76	3.51	31.95	38.60	13.18	16.54	17965
BUD	0.44	0.43	1.45	1.41	3.18	3.03	59.26	60.57	32.27	32.14	9911
GRIG	0.11	0.11	1.81	1.81	3.22	3.21	53.59	53.58	29.64	29.72	8694
TRE	1.39	1.28	2.14	2.04	3.51	3.37	38.21	40.10	18.71	19.55	13499
LOC	-1.53	-1.45	1.99	1.89	4.11	4.06	39.32	41.84	18.23	19.16	10418
BRAS	-1.02	-0.83	1.63	1.59	3.38	3.37	46.96	49.91	22.11	25.07	18083
WWO	0.42	0.43	1.52	1.53	3.19	3.22	60.35	60.30	35.48	35.14	4698
HYM	-2.11	-2.12	1.84	1.78	3.64	3.58	33.10	33.15	14.44	13.99	14207
WCK	2.69	2.72	1.39	1.43	4.00	4.10	31.11	29.91	11.87	11.74	7869
WUS	-1.12	-1.11	1.95	1.84	3.62	3.56	45.03	47.22	23.81	24.97	11187
KIT	1.44	1.31	1.25	1.26	3.08	3.09	53.62	55.22	28.30	30.14	5703

Station code	μ [K]		σ [K]		RMSD [K]		Threshold [%]		Goal [%]		#matchups
	V3.0	V_{PREV}	V3.0	V_{PREV}	V3.0	V_{PREV}	V3.0	V_{PREV}	V3.0	V_{PREV}	V3.0 & V_{PREV}
GIN	-0.48	-0.68	1.26	1.34	3.19	3.19	60.22	56.72	33.58	28.73	35382
GWW	-0.49	-0.86	1.71	1.69	3.86	3.38	49.57	45.54	26.90	22.24	23511
ASM	0.77	0.25	1.28	1.23	3.42	3.17	65.46	66.36	44.91	46.13	24375
ROB	-0.88	-0.96	1.46	1.65	3.03	3.12	61.42	56.49	36.66	33.52	23336
CAL	-0.37	-1.02	1.77	1.73	4.18	4.08	50.99	44.42	30.76	21.48	36771
WHR	-1.03	-1.01	1.32	1.12	3.56	2.77	53.79	61.12	29.45	32.73	9946
YAN	2.13	1.69	2.27	2.18	3.84	3.62	33.30	36.78	16.43	18.86	35210
CBL	-1.14	-1.45	1.44	1.50	3.14	3.20	50.42	44.93	23.34	19.18	29012
GOB	-0.79	-1.42	1.12	1.33	2.38	3.00	65.50	46.43	31.32	18.52	52446
TAT	1.92	1.48	2.88	2.87	3.98	3.85	35.10	33.41	17.33	16.11	22348
total	**-0.429** (±0.007)	**-0.589** (±0.007)	**1.765** (±0.006)	**1.811** (±0.006)	**3.244** (±0.006)	**3.231** (±0.006)	**53.41**	**51.31**	**29.07**	**27.30**	**781375**
total*	**-0.502** (±0.006)	**-0.749** (±0.007)	**1.795** (±0.005)	**1.865** (±0.005)	**3.397** (±0.007)	**3.398** (±0.006)	**52.56**	**49.62**	**28.62**	**26.40**	**835893/ 897919**

The yearly evolution of each performance metric is presented in Figure 4 for both CLMS LST product versions, as well as for the full reprocessing period (last row). An overall underestimation (i.e. negative values) is observed, with V3.0 consistently exhibiting reduced biases throughout the study period in comparison to V_{PREV}. The improvements in V3.0 are particularly evident when considering the impact of algorithm and input updates. The implementation of dynamic vegetation-based emissivity in 2021 for both HMWR and GOES-16 disks and subsequent refinement by the end of 2022, together with the integration of the CLMS SCE product, contribute to a more stable bias behaviour in V3.0 across the following years. In contrast, V_{PREV} only partially benefits from these updates, resulting in a temporal convergence in 2021–2022 that is followed by a renewed degradation. Through 2023 and 2024, V3.0 maintains a more consistent bias (around -0.53 and -0.48 K, respectively), while larger underestimations are observed in V_{PREV} (-0.69 and -0.63 K, respectively). Such difference reflects the correct handling of emissivity coefficients in V3.0 following the transition from HMWR-8 to HMWR-9, ensuring improved consistency over the entire 8-year period. Regarding the absolute residuals and RMSDs, differences between both versions are smaller. V3.0 shows consistently lower deviations (between 1.68–1.82 K) compared to V_{PREV} (between 1.76–1.85 K). RMSD values remain generally comparable with, however, higher values during 2021–2023. This is probably due to some GBOV stations in those years that were affected by an increased number of cloudy pixels. Over the full study period, V3.0 maintains improved overall performance, despite a marginal increase of +0.01 K in RMSD relative to V_{PREV}.

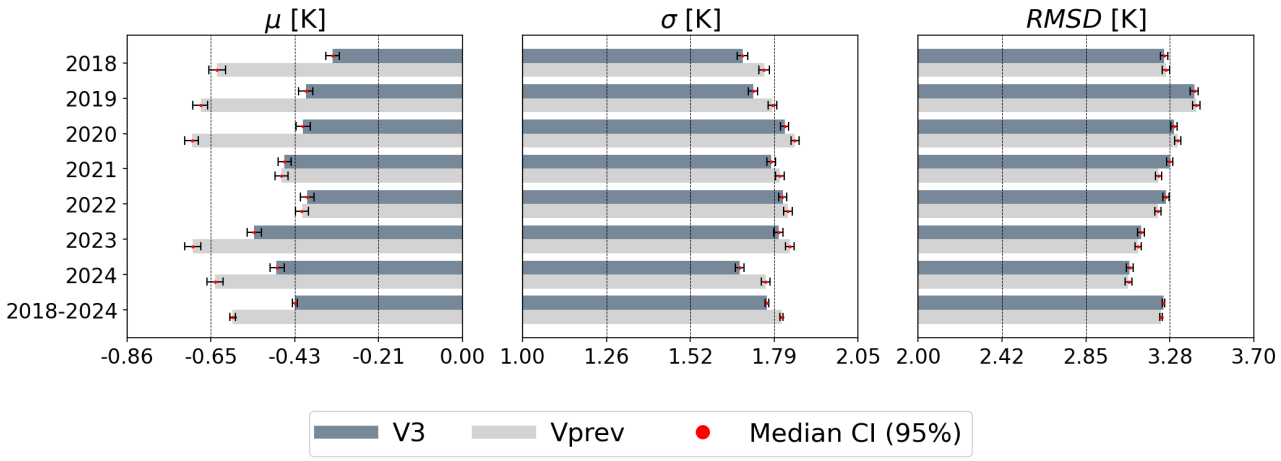


Figure 4 - Statistical summary of yearly metrics – bias (μ), absolute residuals (σ), and root-mean-squared difference (RMSD) – derived from all hourly satellite/in-situ matchups in respect to CLMS LST V3.0 and V_{PREV} over all GBOV stations considering the same matchups. Statistics for the full study period (2018–2024) are shown in the last row, including the 95 % confidence intervals relative to the median of each metric. All values are expressed in Kelvin.

Figure 5 shows density plots of all hourly satellite/in-situ matchups from January 2018 to December 2024 for two example GBOV stations: BON and PSU. For each site, results from V3.0 (left column) and V_{PREV} (right column) are shown side-by-side. The density representation highlights the concentration of matchup points in relation to the 1:1 line and allows a visual assessment of how each CLMS LST version performance varies over different surface temperature ranges. In particular, these plots allow to identify systematic tendencies towards over-or underestimations at low and high LSTs, complementing the previous statistical analysis.

In station BON, both versions show a comparable overall performance, with V3.0 exhibiting a lower bias (-0.58 K) associated to a better alignment of lower temperatures along the 1:1 line. During winter, this site can experience snow cover, and, as observed in other snow-affected stations, V3.0 is set to provide more accurate LST estimates than V_{PREV} due to refined snow cover input, and therefore better representation of cold surface temperatures under such conditions. In contrast, for higher temperatures, V3.0 shows somewhat greater dispersion than V_{PREV}, with a few additional outliers that lead to higher median absolute residuals and RMSD values (i.e. increases of +0.02 K and +0.04 K, respectively). At station PSU, V3.0 shows better performance metrics in comparison to V_{PREV}, with the main density cluster more closely aligned with the 1:1 line. A low positive bias (0.07 K) and an RMSD of 1.98 K are obtained for V3.0, while V_{PREV} presents larger bias (0.17 K), RMSD (2.78 K), and absolute residuals (1.91 K). V3.0 corrects the large dispersion observed in V_{PREV}, particularly at lower temperatures, where the latter shows a systematic overestimation under snow conditions. Improvements are also evident at higher temperatures (> 300 K), where the underestimation observed

in V_{PREV} is corrected in V3.0, leading to a better alignment with the 1:1 line. For intermediate temperatures (260–300 K), V3.0 exhibits a narrower distribution, indicating improved consistency across the temperature range.

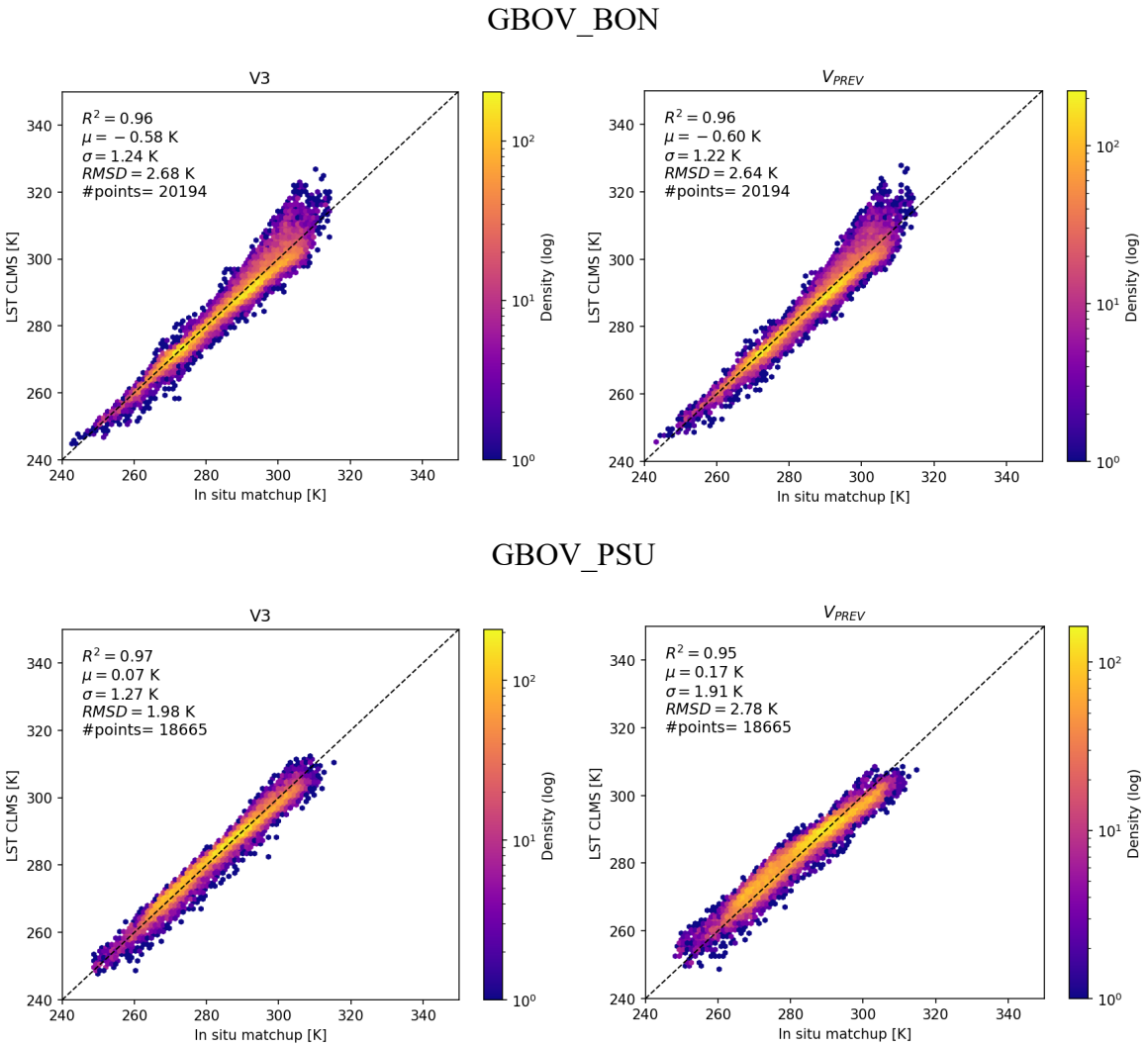


Figure 5 - Density plots comparing CLMS LST V3.0 (left column) and V_{PREV} (right column) hourly matchups (in Kelvin) to in-situ observations (2018–2024) for GBOV stations: Bondville (GBOV_BON, top), and Rock Springs (GBOV_PSU, bottom). The corresponding performance metrics are also added: bias (μ), absolute residuals (σ), and root-mean-squared difference (RMSD).

5.1.3 Monthly variations of the statistics

The monthly evolution of each performance metric is shown in Figure 6 for two GBOV stations: Williams (USA, GOES-16 disk) and Gingin (Australia, HMWR disk). Two consecutive years (2020

and 2021) are selected to showcase the implementation of the dynamic vegetation cover 1 km on January 2021 in V_{PREV} when compared to V3.0. For both stations, monthly statistics reveal a clear change of behaviour associated with the introduction of dynamic vegetation. In particular, V_{PREV} metrics become more in line with the ones found with V3.0, especially the monthly biases.

At station Williams, V3.0 exhibits a more consistent temporal behaviour throughout the entire period, with reduced bias amplitude and lower RMSD values compared to V_{PREV}. In 2020, V3.0 shows biases between ±1 K, while larger deviations are observed in V_{PREV} (down to -2 K in June). The largest discrepancies occur during summer months when reduced vegetation cover enhances surface heterogeneity; however, V3.0 maintains lower RMSD values (about 3.5 K compared to 4 K), indicating a more robust representation of such surface conditions. The stable behaviour in V3.0 reflects the consistent use of dynamic vegetation information across the entire period, allowing more coherent seasonal estimates of LST. Consequently, the temporal variability of V3.0's metrics remain smoother and more physically consistent, while V_{PREV} shows stronger fluctuations prior to the implementation of such updates.

At Gingin station, V3.0 similarly shows improved consistency, with systematically lower biases and generally reduced RMSDs over 2020. Underestimations during austral winter are reduced (about -0.5 K compared to -1.2 K in V_{PREV}), and seasonal transitions are better represented. During warmer months, RMSD increases due to stronger thermal contrasts (e.g. sunlit versus shadowed ground, vegetation, etc.), however V3.0 maintains generally lower values relative to V_{PREV}. In 2021, following the introduction of dynamic vegetation cover in January, the temporal evolution of the metrics becomes more consistent between both versions, showing minimal discrepancies and better alignment of all metrics curves throughout the year. The stability of V3.0 is further highlighted in 2024, where it maintains consistent performance due to the correct treatment of emissivity coefficients throughout the entire period, which is in agreement with the cross-comparison of V3.0 and V_{PREV} time-series presented in Figure 3.

These results indicate that the continuous representation of dynamic vegetation provides a more stable temporal behaviour of the CLMS product, allowing a realistic seasonal evolution of LST estimates that aligns with local land cover variability. Overall, V3.0 delivers a more coherent and physical consistent temporal LST record than V_{PREV} over the entire study period.

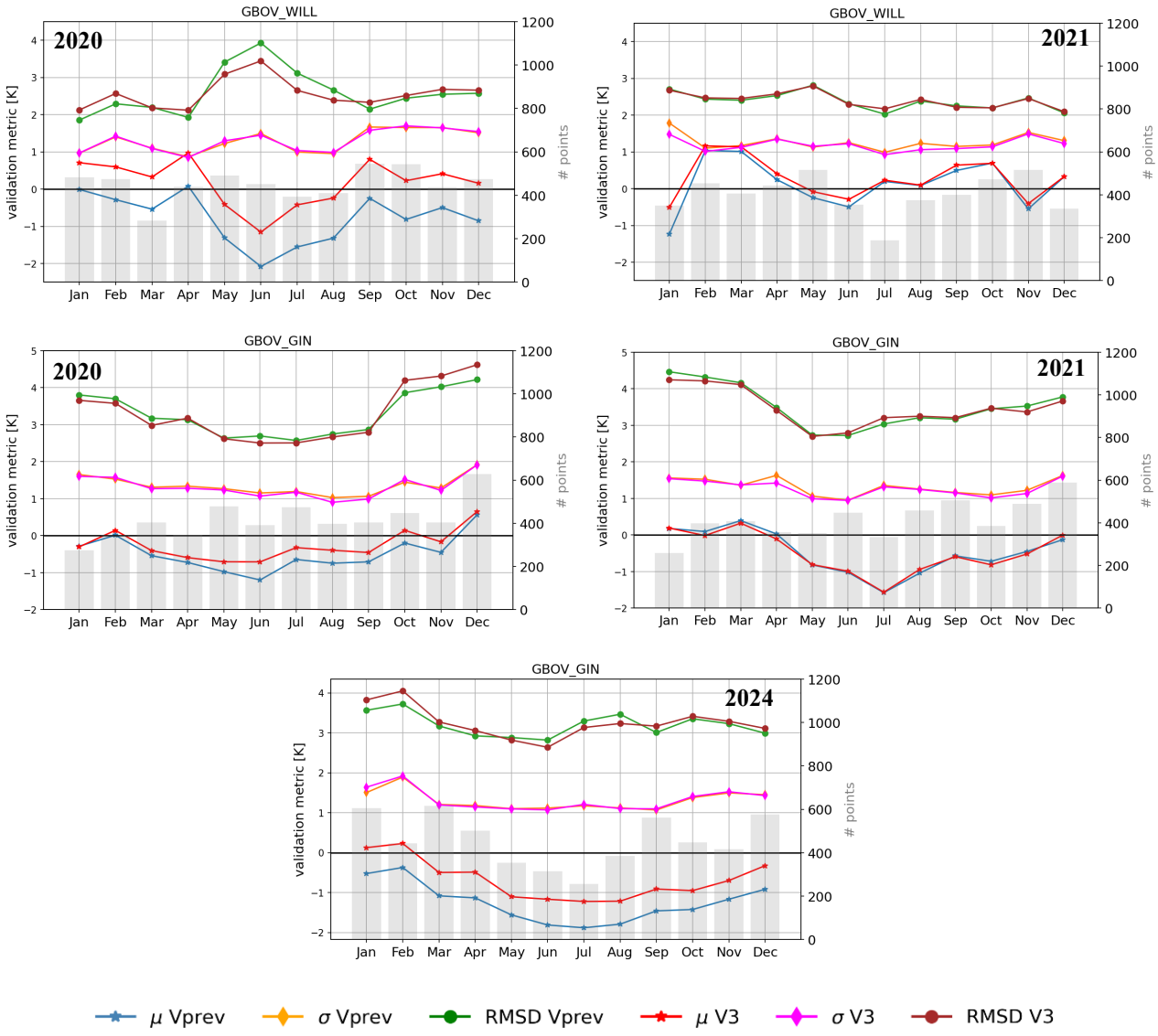


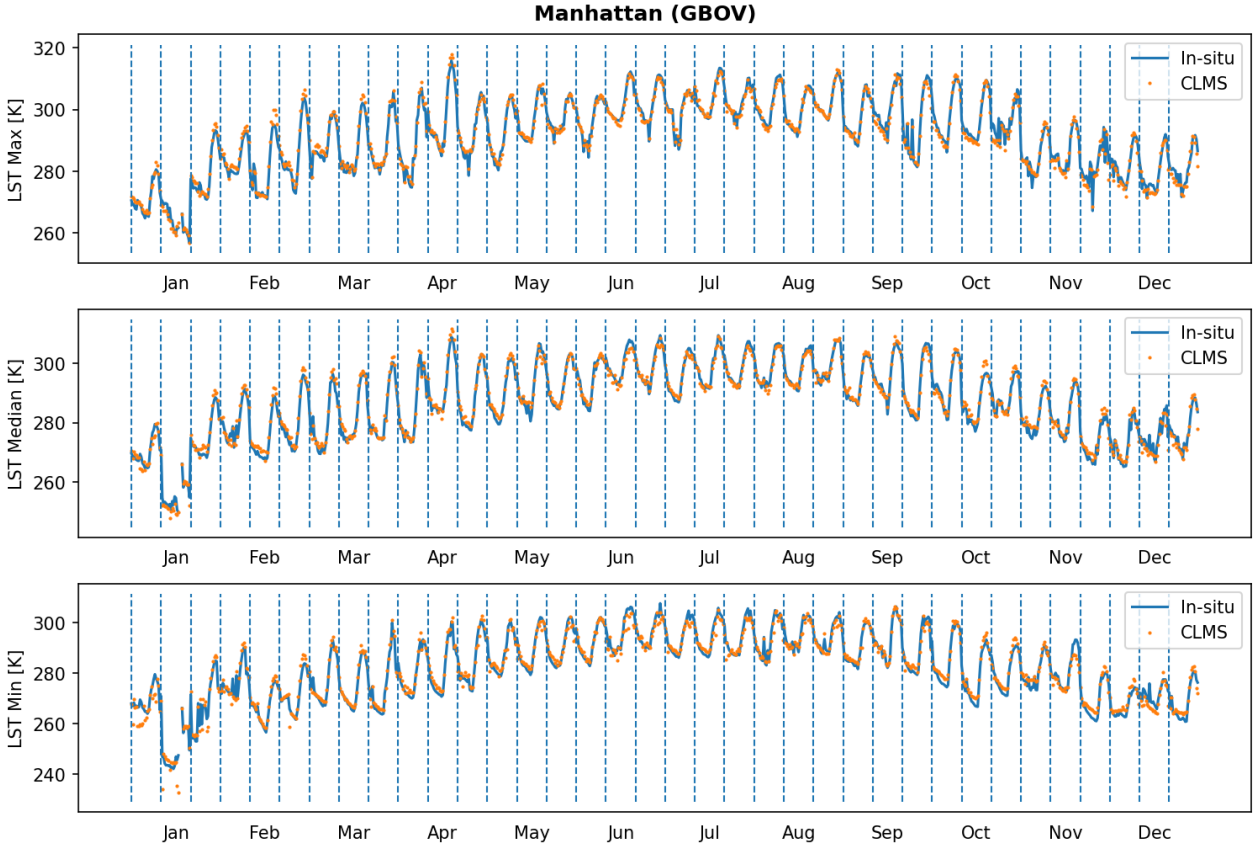
Figure 6 - Monthly statistical metrics between CLMS LST V3.0 and V_{PREV} for 2020 (left column) and 2021 (right column) at GBOV stations: Williams (GBOV_WILL, top) and Gingin (GBOV_GIN, bottom). The 2024 metrics are also shown for GBOV_GIN. Statistical metrics include: bias (μ), absolute residuals (σ), and root-mean-squared difference (RMSD). Grey bars indicate the number of available matchups (right-hand axis). All values are expressed in Kelvin.

5.1.4 Results for LST10-DC products

The LST10-DC family of products describes the properties of the diurnal cycle of LST during a dekad. The maximum (Max), minimum (Min), and median (Med) LST values for each hour are derived for each dekad from hourly LSTs, being then compared to the corresponding statistics derived from in-situ data. In Figure 7, a representation is provided as an example for the 2024 mean dekad

diurnal cycle of GBOV_MANH and GBOV_DRA stations generated by the CLMS LST10-DC V3.0 product. As in the LST10 Quality Assessment Report [CGLOPS1_QAR_LST-LST10-V2.1], in-situ values within a 1-hour time window around each cloudy satellite observation were masked out, so that the statistics derived from observations are representative of clear-sky conditions.

For GBOV_MANH, the main features of the diurnal cycle are generally well reproduced. Differences are observed with small overestimations of V3.0 around midday during winter, spring and autumn. Additional deviations occur during nighttime (i.e. the initial part of each section represented within two vertical dashed lines), particularly for minimum temperatures between September and December. This station is located near large crop fields, where temporal changes in vegetation cover can affect the representativeness of the pixel-scale satellite observations. Periods of larger discrepancies are also associated with irregular diurnal curves, suggesting reduced sampling due to cloud cover.



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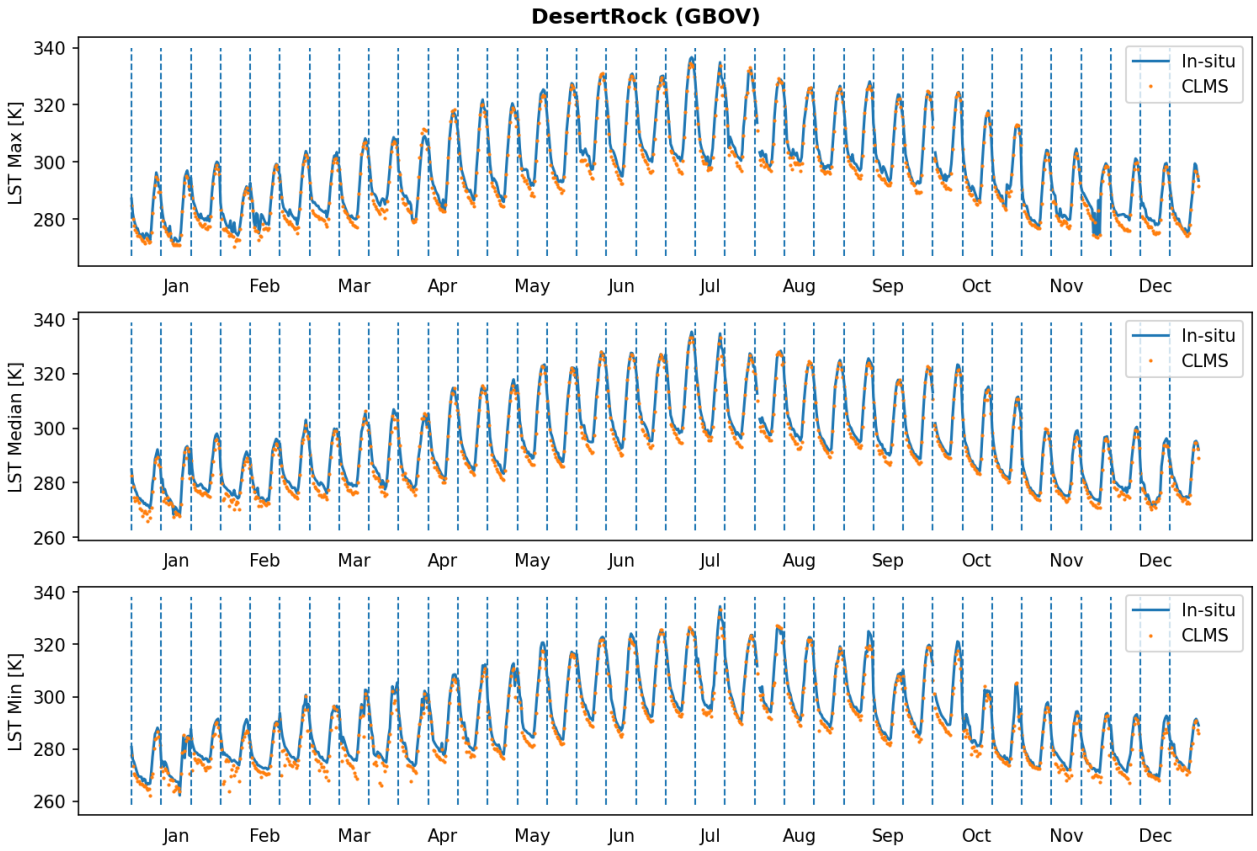


Figure 7 - Representation of the LST10-DC product for Manhattan (GBOV_MANH) and Desert Rock (GBOV_DRA) stations for 2024. Each section in between the vertical dashed lines represents a dekadal mean diurnal cycle. The top panel shows LST Max; LST Median is shown in the middle and LST Min is shown in the bottom panel of each set. The blue curves represent the observations statistics, while the orange dots represent CLMS LST V3.0. All values are expressed in Kelvin.

For GBOV_DRA, the overall temporal behaviour remains consistent with in-situ observations throughout the year. Differences are mainly observed during nighttime, with lower satellite-derived LSTs relative to in-situ measurements, especially for minimum temperatures in winter. This station is located in a heterogeneous environment, near an airport and is surrounded by shrubland, hills, and rocky surfaces. This spatial variability in surface properties contributes to known representativeness issues when comparing satellite and in-situ LSTs, as discussed in section 5.1.2.

Scatterplots comparing CLMS LST10-DC V3.0 values for all timeslots and the corresponding estimates using in-situ data are shown in Figure 8 for stations GBOV_SXF and GBOV_GWN over the 2024 period. In the case of GBOV_SXF, higher temperatures are overestimated in the Max, Median, and Min products, while underestimation occurs for lower temperatures, particularly in the Min metric. The Median value product (LST10-DCMed) is reasonably well-estimated with a

coefficient of determination (R^2) of 0.98 and RMSD of 2.09 K, while both Max (LST10-DCMax) and Min (LST10-DCMin) products are shown to have relatively higher RMSDs (2.27 and 2.60 K, respectively) and practically the same alignment towards in-situ (R^2 = 0.97). For GBOV_GWN, higher errors are obtained, particularly for the Min metric (RMSD = 3.13 K), where minimum temperatures are generally overestimated (μ = 0.44 K) and maximum temperatures are underestimated (μ = -0.44 K).

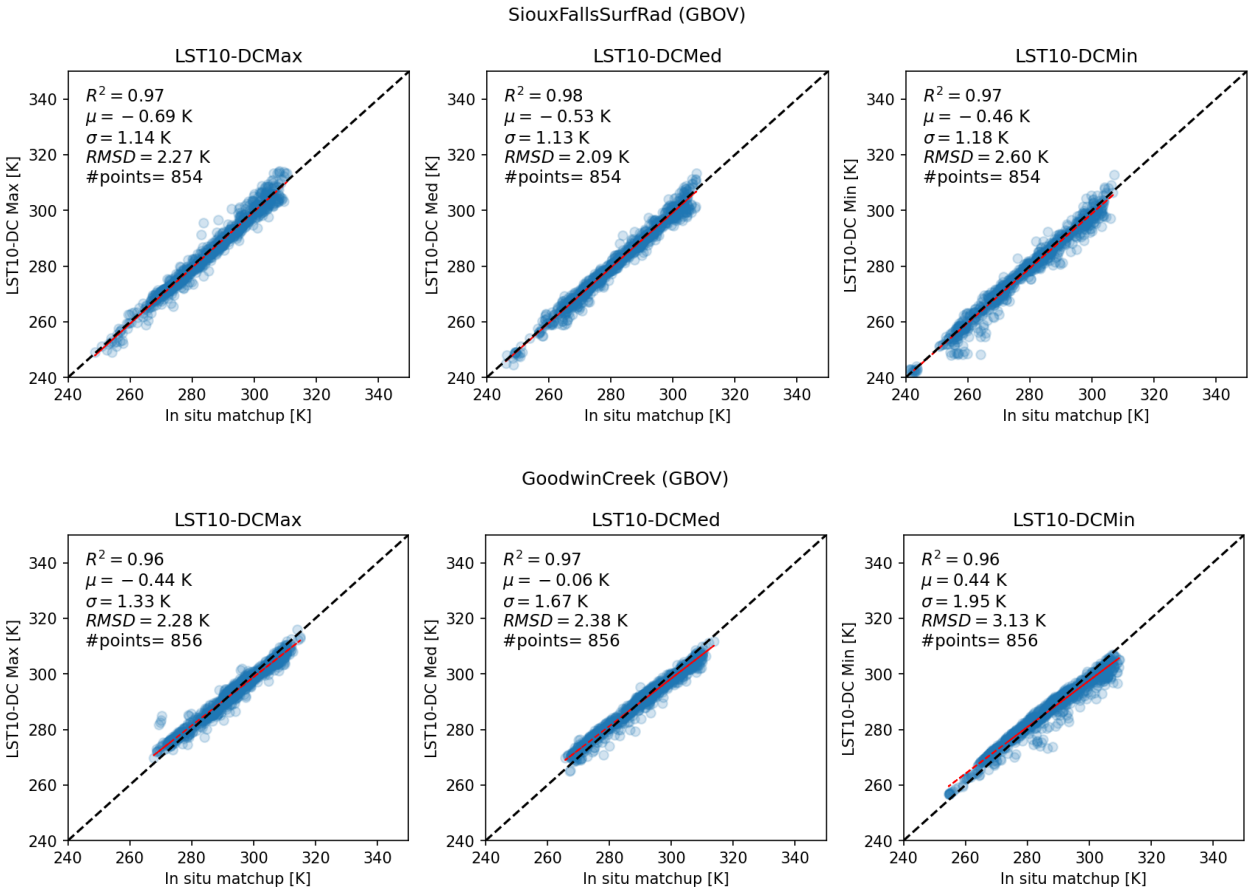


Figure 8 - Scatterplots for the comparison of CLMS LST10-DC V3.0 with the respective in-situ estimates in Sioux Falls (GBOV_SXF) and Goodwin Creek (GBOV_GWN) stations. Estimates for the Max (left), Median (middle) and Min (right) statistics computed over each dekad are compared for all timeslots in 2024. Only clear-sky situations were used for in-situ data.

Summary statistics are provided for the maximum, median, and minimum LSTs daily cycle (Table 3, Table 4, and Table 5, respectively). For the LST10-DC Max (Table 3), V3.0 shows smaller biases, standard deviations, and RMSD values in comparison with V_{PREV} results across all years, as well as considering the entire study period. For instance, for the latter case and considering the same number

of satellite/in-situ matchups (2018–2024), i.e. 74849 samples for both versions, V3.0 exhibits a median bias of about -0.57 K, standard deviation of about 1.70 K, and a RMSD of about 3.26 K, while V_{PREV} shows a larger bias (about -1.20 K), standard deviation (about 1.81 K), and RMSD (about 3.55 K). When considering each dataset with the original number of matchups over the full period (2018–2024*), i.e. 170771 for V3.0 and 74940 for V_{PREV}, the new V3.0 version continues to maintain lower errors despite having almost the double number of samples than the operational.

Table 3 – Statistical summary of yearly metrics – bias (μ), absolute residuals (σ), and root-mean-squared difference (RMSD) considering corresponding LST10-DC Max satellite/in-situ matchups from all GBOV stations for CLMS LST V3.0 and V_{PREV}. Statistics for the full study period with same matchups for V3.0 and V_{PREV} (2018–2024) and with independent matchups (2018–2024*) are shown in the last two rows, including the 95 % confidence interval between brackets. All values are expressed in Kelvin.

Period	μ [K]		σ [K]		RMSD [K]		#matchups	
	V3.0	V_{PREV}	V3.0	V_{PREV}	V3.0	V_{PREV}	V3.0	V_{PREV}
2018	-0.47	-1.25	1.66	1.84	3.37	3.74	9155	9155
2019	-0.55	-1.24	1.72	1.86	3.57	3.87	9479	9479
2020	-0.59	-1.36	1.77	1.89	3.42	3.75	12103	12103
2021	-0.62	-1.03	1.65	1.76	3.23	3.39	12258	12258
2022	-0.57	-1.01	1.72	1.82	3.20	3.48	11873	11873
2023	-0.60	-1.23	1.75	1.80	3.06	3.31	10772	10772
2024	-0.62	-1.31	1.60	1.67	2.90	3.30	9209	9209
2018–2024	**-0.574** (±0.020)	**-1.199** (±0.021)	**1.697** (±0.015)	**1.806** (±0.017)	**3.255** (±0.022)	**3.549** (±0.021)	**74849**	**74849**
2018–2024*	**-0.514** (±0.015)	**-1.199** (±0.021)	**1.789** (±0.011)	**1.806** (±0.017)	**3.350** (±0.016)	**3.551** (±0.021)	**170771**	**74940**

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Table 4 – Same as Table 3 but for LST10-DC Med.

Period	μ [K]		σ [K]		RMSD [K]		#matchups	
	V3.0	VPREV	V3.0	VPREV	V3.0	VPREV	V3.0	VPREV
2018	-0.46	-0.93	1.67	1.73	3.23	3.21	9155	9155
2019	-0.46	-0.94	1.69	1.72	3.36	3.38	9479	9479
2020	-0.52	-0.98	1.78	1.75	3.17	3.18	12103	12103
2021	-0.61	-0.78	1.70	1.64	3.07	3.03	12258	12258
2022	-0.51	-0.68	1.70	1.72	2.99	2.94	11873	11873
2023	-0.58	-0.91	1.75	1.79	2.96	2.96	10772	10772
2024	-0.48	-0.85	1.65	1.75	2.89	2.93	9209	9209
2018–2024	**-0.523** (±0.020)	**-0.860** (±0.022)	**1.707** (±0.016)	**1.728** (±0.016)	**3.095** (±0.020)	**3.088** (±0.019)	**74849**	**74849**
2018–2024*	**-0.542** (±0.015)	**-0.860** (±0.022)	**1.794** (±0.011)	**1.728** (±0.016)	**3.204** (±0.015)	**3.088** (±0.019)	**170771**	**74940**

The statistical summary for LST10-DC Med (Table 4) highlights the behaviour of the most relevant metric within the LST10-DC family of products. The median LST is less sensitive to extreme values than the maximum and minimum and therefore provides a better representation of the dataset performance. In terms of bias, V3.0 consistently shows smaller values than V_{PREV} for each individual year as well as over the entire study period. When considering the same number of samples, both versions exhibit an overall underestimation with respect to the in-situ observations, with V3.0 showing a lower median bias (about -0.52 K) compared with V_{PREV} (-0.86 K). When the original number of samples is considered, V3.0 continues to show lower biases, despite a slight increase in magnitude resulting from the substantial increase in the number of matchups (170771) compared with V_{PREV} (74940). In contrast, V_{PREV} produces nearly the same results since the number of samples changes only marginally. Regarding the other metrics, the discrepancies between the two datasets are small. The standard deviation for V3.0 is higher in 2020 and 2021 compared to V_{PREV} (+0.03 and +0.06 K, respectively), while for the remaining periods V3.0 exhibits lower deviations, including the entire study period (about -0.02 K relative to V_{PREV}). Similarly, the RMSD for V3.0 is marginally higher in 2021 and 2022 (+0.04 and +0.05 K, respectively) relative to V_{PREV}, equal in 2023 (2.96 K),

and very close for the full period when using the same number of samples (about 3.10 and 3.09 K, respectively). When considering the original number of samples, V3.0 shows an RMSD increase of +0.12 K relative to V_{PREV}. Despite these small discrepancies between the CLMS LST product versions for the median LST values, these remain minor, indicating that V3.0 consistently produces 10-daily LST estimates that are generally comparable to the operational dataset, while still benefiting from reduced biases and a substantially larger number of valid matchups. The latter is mainly related to differences in the product version processing approach: V_{PREV} had a more restrictive processing constraints, which limit the number of valid retrievals, while these constraints have been removed in V3.0 reprocessing, allowing a larger number of observations to be retained, leaving the decision to filter data based on the fraction of available data to each user and application.

Table 5 – Same as Table 3 but for LST10-DC Min.

Period	μ [K]		σ [K]		RMSD [K]		#matchups	
	V3.0	V_{PREV}	V3.0	V_{PREV}	V3.0	V_{PREV}	V3.0	V_{PREV}
2018	-0.51	-0.58	1.92	1.99	3.63	3.46	9155	9155
2019	-0.58	-0.58	1.86	1.97	3.63	3.57	9479	9479
2020	-0.51	-0.50	1.98	1.98	3.52	3.33	12103	12103
2021	-0.60	-0.35	1.90	1.89	3.56	3.30	12258	12258
2022	-0.62	-0.33	1.91	1.95	3.45	3.26	11873	11873
2023	-0.67	-0.55	1.89	1.94	3.34	3.24	10772	10772
2024	-0.53	-0.45	1.83	1.98	3.34	3.32	9209	9209
2018–2024	**-0.576** (±0.023)	**-0.469** (±0.024)	**1.905** (±0.019)	**1.956** (±0.019)	**3.497** (±0.024)	**3.350** (±0.022)	**74849**	**74849**
2018–2024*	**-0.644** (±0.018)	**-0.469** (±0.023)	**2.997** (±0.012)	**1.956** (±0.019)	**3.713** (±0.020)	**3.351** (±0.022)	**170771**	**74940**

For the case of LST10-DC Min (Table 5), the statistical summary shows a different behaviour compared with the maximum and median LST daily cycle products. In terms of bias, V3.0 generally exhibits stronger underestimation relative to the in-situ observations than V_{PREV} for most years. Over

the full period, considering the same number of samples, V3.0 shows a bias of about -0.58 K compared with -0.47 K for V_{PREV}, i.e. an increase of -0.11 K. This tendency becomes more pronounced when the original number of samples is considered, with an increase of about -0.18 K, which is likely due to some stations (e.g. DRA, LOC, CAL) that are known to have representativeness issues when comparing satellite and in-situ LSTs, contributing to an increased deviation. For the remaining metrics, the standard deviation is lower for V3.0 than V_{PREV} over the full period when considering the same number of matchups (about 1.91 K versus 1.96 K), although the RMSD is somewhat higher (about 3.50 K versus 3.35 K). When considering the original number of samples, the RMSD discrepancies increase, particularly for V3.0, reflecting the substantial larger number of matchups available. Despite these discrepancies between versions, the results remain within a comparable range to the previous operational dataset.

5.1.5 Results for LST10-Syn products

This family of products characterizes the range of LSTs along a dekad, regardless of the time of day. Similarly to LST10-DC, the maximum, median, and minimum LSTs are derived for each dekad, but with data from all timeslots. For this analysis, the 10-daily maximum and minimum synthesis products are considered auxiliary fields, as these are more susceptible to sampling uncertainties, which makes comparisons with in-situ observations more challenging. The median LST is therefore treated as the primary synthesis product. It is worth noting that since the Thermal Condition Index (TCI) cannot be directly validated with in-situ observations due to the lack of long time series in from the GBOV network, the TCI is indirectly validated via the LST validation.

Examples of time-series from this family of products are shown in Figure 9 for GBOV_DMOS and GBOV_TRE. Only one value of each metric is derived per dekad, which results in significantly fewer missing satellite observations compared to the hourly and LST10-DC products. For GBOV_DMOS, there is generally good agreement between the satellite estimates (in orange) and the station measurements (in blue). Some dekads show satellite underestimation, which in most cases can be attributed to cloud cover. Similarly, for GBOV_TRE, small underestimations occur during winter, while overestimations take place during warmer periods. These two examples show the usefulness of the LST10-Syn products in synthetically describing the seasonal cycle of LST at the pixel level.

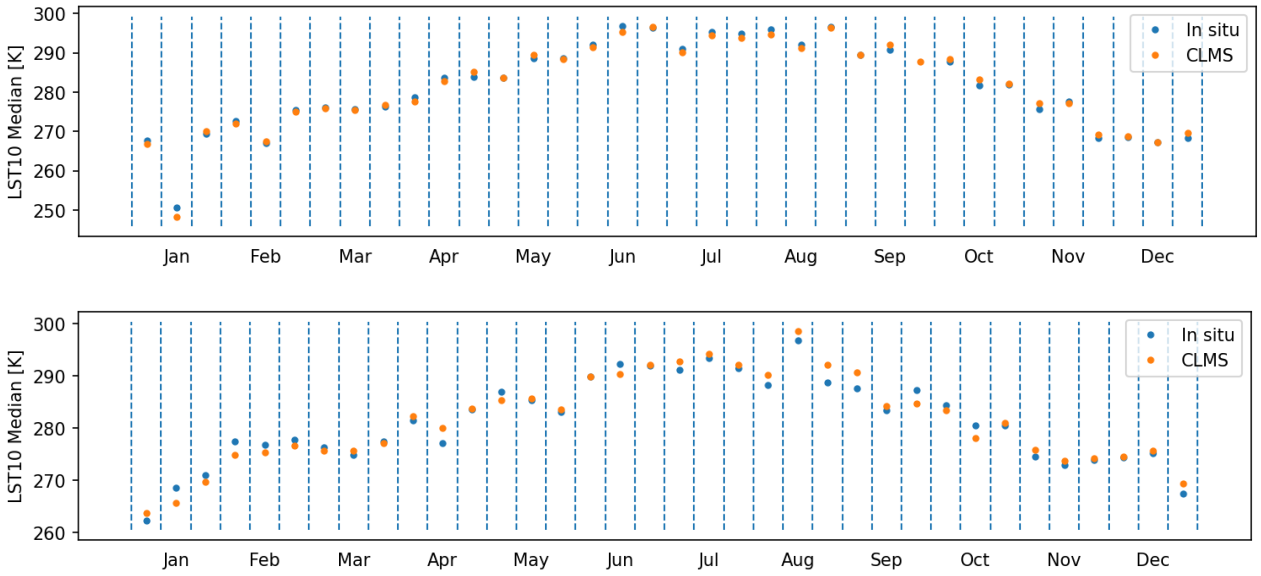


Figure 9 - Time-series of the LST10-Synthesis median products for Des Moines (GBOV_DMOS, top) and Trebon (GBOV_TRE, bottom) stations for 2024. Each section in between the vertical dashed lines represents a dekadal period. The blue curves represent the observations statistics, while the orange dots represent CLMS LST V3.0. In-situ statistics were derived with clear-sky situations only. All values are expressed in Kelvin.

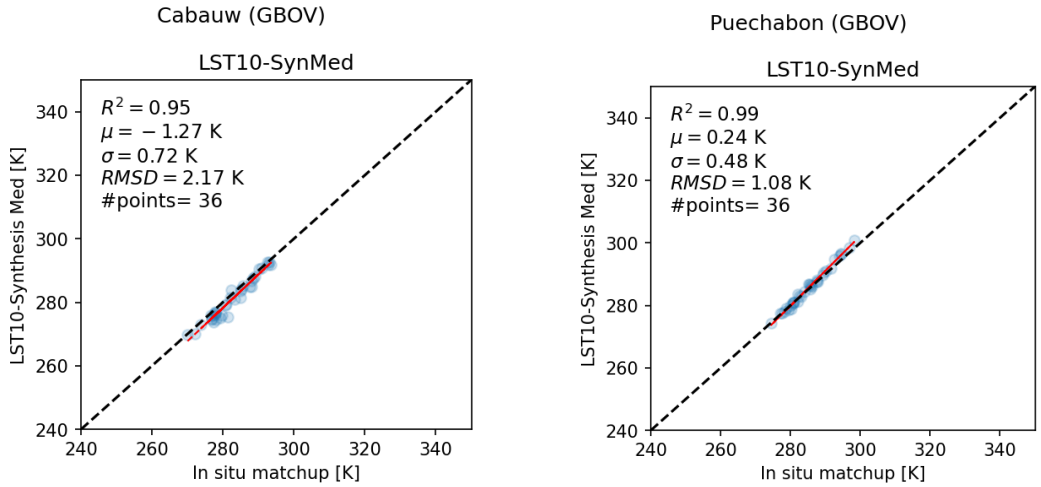


Figure 10 - Scatterplots for the comparison of CLMS LST10-Synthesis median (Med) V3.0 with the respective in-situ estimates in Cabauw (GBOV_CAB, left) and Puechabon (GBOV_PUEC, right) stations. Estimates Median statistics computed over each dekad are compared for all timeslots in 2024. Only clear-sky situations were used for in-situ data.

Figure 10 presents two examples of scatterplots comparing each LST10-Syn median product with the corresponding clear-sky in-situ statistics. It is worth noting that sampling is significantly reduced relative to the previous evaluations, with only 36 dekads per year. In the two examples shown (GBOV_CAB and GBOV_PUEC), the satellite LST10-Syn median values present an overall agreement with in-situ observations, highlighting the capability of this synthesis product to capture the LST seasonality. For GBOV_CAB, satellite LST estimates usually underestimate in-situ values for the lower temperature range, while agreement is generally closer for mid-higher temperatures. For GBOV_PUEC, small overestimations are observed for high temperatures, while maintaining a better alignment along the 1:1 line for the mid-lower temperature range.

Table 6 - Same as Table 3 but for LST10-Syn Med.

Period	μ [K]		σ [K]		RMSD [K]		#matchups	
	V3.0	VPREV	V3.0	VPREV	V3.0	VPREV	V3.0	VPREV
2018	-0.84	-1.14	1.10	1.06	2.22	2.40	879	879
2019	-1.06	-1.38	1.14	1.13	2.39	2.46	992	992
2020	-0.99	-1.35	1.21	1.20	2.26	2.35	1178	1178
2021	-1.04	-1.14	1.23	1.18	2.73	2.83	1221	1221
2022	-0.82	-0.92	1.21	1.18	2.15	2.17	1201	1210
2023	-1.01	-1.34	1.26	1.11	2.17	2.18	1002	1002
2024	-0.93	-1.35	1.19	1.15	2.13	2.26	895	895
2018–2024	**-0.950** (±0.058)	**-1.240** (±0.054)	**1.196** (±0.030)	**1.153** (±0.029)	**2.315** (±0.069)	**2.397** (±0.082)	**7368**	**7368**
2018–2024*	**-0.953** (±0.058)	**-1.240** (±0.054)	**1.198** (±0.031)	**1.155** (±0.029)	**2.333** (±0.069)	**2.398** (±0.083)	**7436**	**7368**

Summary statistics for LST10-Syn are shown only for the median in Table 6. As previously mentioned, the synthesis maximum and minimum over a 10-day period are prone to significant sampling uncertainties, making their interpretation and comparison with in-situ observations difficult. For the LST10-Syn Med, V3.0 consistently shows smaller biases than VPREV across all years and full

period. In the latter, considering the same number of matchups, V3.0 exhibits a median bias of -0.95 K compared with -1.24 K for V_{PREV}, with similar values being obtained when using independent samples. Standard deviations for V3.0 and V_{PREV} are similar, with a reduction of the RMSD in V3.

5.2 SPATIAL CONSISTENCY AND TEMPORAL STABILITY

5.2.1 Spatial consistency: V3.0 vs V_{PREV}

The consistency of the spatial distributed patterns between CLMS LST product V3.0 and V_{PREV} is assessed qualitatively in Figure 11 through the comparison of seasonal mean difference maps. Seasonal mean differences between V3.0 and V_{PREV} are analyzed for two separate periods (2018–2020 and 2021–2025), corresponding to important updates in the operational CLMS LST processing chain through the study period, while the seasonal outputs are computed for December–January–February (DJF) and for June–July–August (JJA) within each corresponding period.

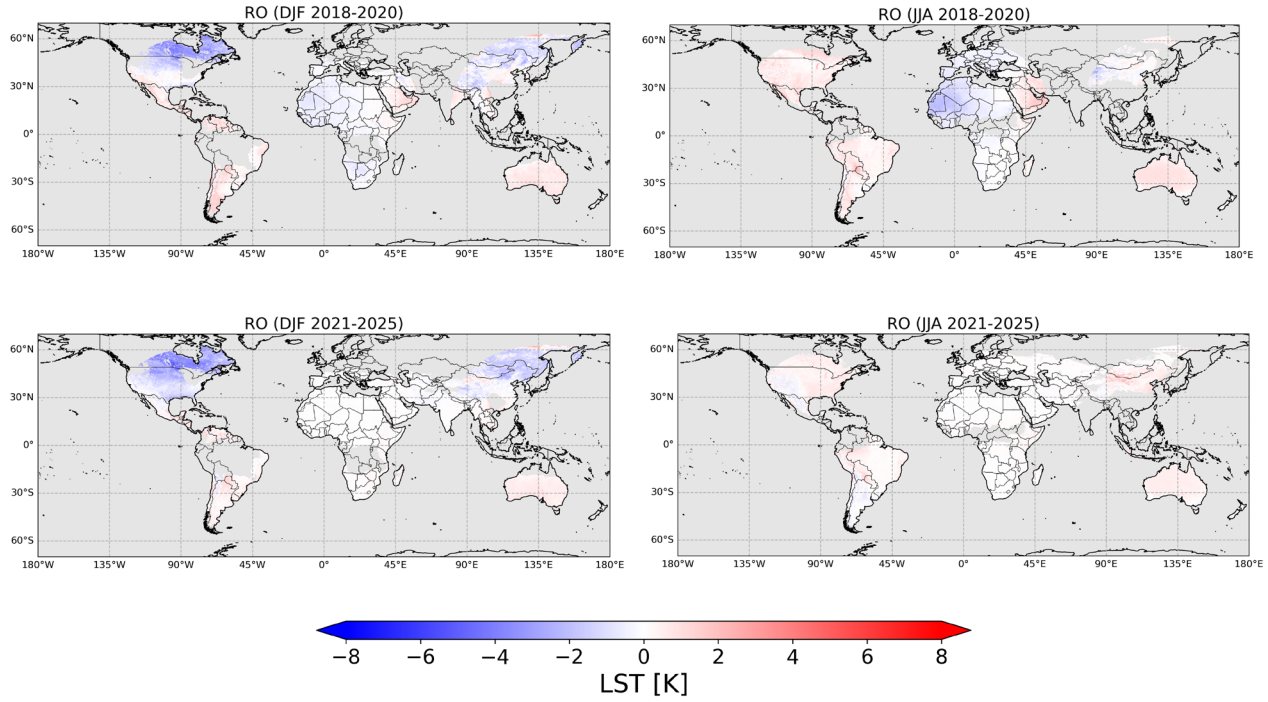


Figure 11 – Seasonal mean LST differences (in Kelvin) between CLMS LST reprocessed V3.0 and previous operational (RO) for December–January–February (DJF) (left column) and June–July–August (JJA) (right column) for two separate periods: 2018–2020 (top) and 2021–2025 (bottom).

For the 2018–2020 period, large discrepancies are observed over the Western Sahara during summer, where V3.0 LSTs are colder (i.e. negative differences) than V_{PREV}. This contrast is significantly reduced in the following years (2021–2025), showing the positive impact of the incremental updates introduced in the CLMS LST retrievals, particularly with the integration of the IODC data in January 2021. For the 2021–2025 summer period, similar patterns are observed over North and South America and Australia, with the latter also being influenced by the transition to HMWR-9. In winter, V3.0 LSTs generally remain colder than the ones produced by V_{PREV} over higher latitudes (e.g. eastern USA, Canada, and northern China), while remaining warmer in the south hemisphere (e.g. Australia, South America) – a consistent behaviour over both analyzed periods. After the implementation of the IODC data, the discrepancies between versions are significantly reduced over the MSGs domain.

5.2.2 Temporal stability with ERA5 reanalysis

Figure 12 presents the monthly mean anomaly differences for mean LST over the global and individual GEO domains, while the corresponding results for maximum and minimum LST are provided in Figure A2 and Figure A3 in the Appendix. Overall, the CLMS LST V3.0 exhibits enhanced temporal stability in comparison to V_{PREV}, as indicated by systematically smaller anomaly differences with respect to ERA5. At the global scale, mean LST anomaly differences for V3.0 range within ±0.2 K until the end of 2022. From 2023 onwards, differences become predominantly negative, reaching values close to -0.4 K during the second half of 2023. This period coincides with the MSG4 – MSG3 transition, being noticeable in the MSG 0º + IODC disk. A marginal increase towards small differences is observed throughout 2025. The increased stability in V3.0 global dataset is further reflected in its lower trend (-0.15 K per decade) and reduced variability (total standard deviation of 0.12 K), in comparison to V_{PREV} (-0.25 K per decade and 0.16 K, respectively).

This behaviour is consistently observed over the individual GEO domains. For mean LST, V3.0 exhibits significantly smaller anomaly difference trends than V_{PREV}, with values of -0.02, -0.16, and -0.19 K per decade for GOES-16, MSG 0º + IODC, and HMWR, respectively, against values of 0.34, -0.57 and -0.40 for V_{PREV}. The reduced trend values are complemented by lower variability (0.14, 0.17, and 0.15 K, respectively), indicating better temporal consistency in the reprocessed dataset. For the MSG 0º + IODC domain, anomaly differences from V3.0 and V_{PREV} converge from 2021 onwards. This approximation is consistent with the introduction of dynamic vegetation at 1 km and the IODC recalibration in the operational version in January 2021. The increase of negative anomalies observed during the second half of 2023 is noticeable in both datasets, with values of about -0.38 K for V3.0 and -0.45 K for V_{PREV}. Over the GOES-16 and HMWR domains, several major LST algorithm updates contributed to the existence of temporal discontinuities in V_{PREV}. These include the introduction of dynamic vegetation-based surface emissivity at 1 km in January 2021, the subsequent upscale to 300 m, and the implementation of the CLMS SCE at 1 km product for emissivity estimation in December 2022, which are already considered in V3.0 throughout the entire

reprocessing period. Moreover, for the HMWR domain, following the transition from HMWR-8 to HMWR-9 in December 2022, V_{PREV} continued to use inconsistent emissivity coefficients, resulting in larger negative anomaly differences in relation to ERA5 in subsequent years – a behaviour not observed in V3.0.

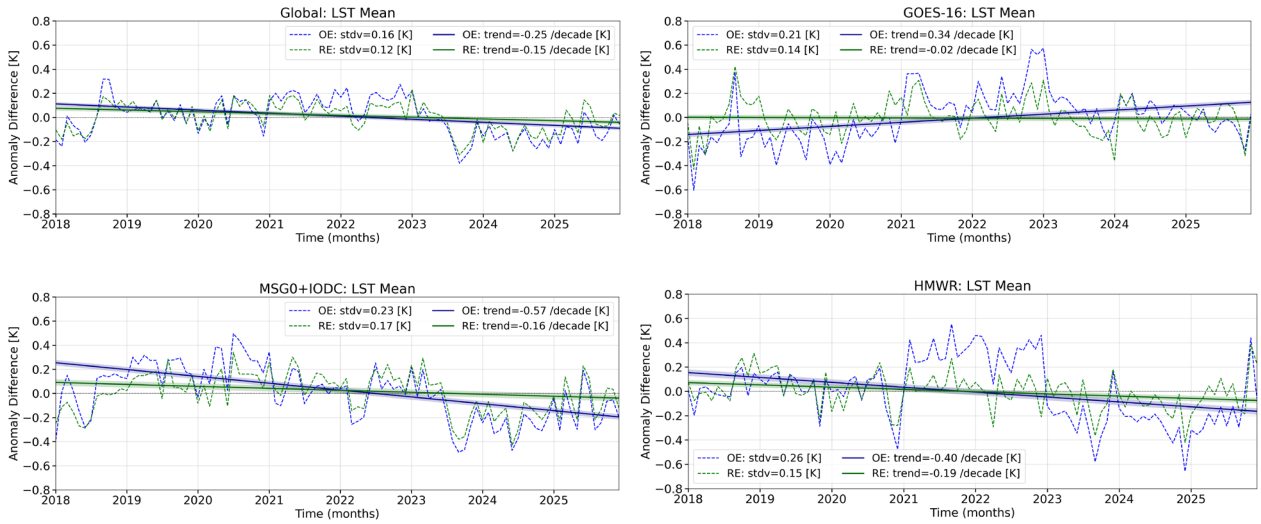


Figure 12 - Monthly mean LST anomaly differences (in Kelvin) between CLMS LST operational version and ERA5 (OE) and between CLMS LST reprocessed version and ERA5 (RE) from January 2018 to December 2025 for separate disk domains: Global (top left); GOES-16 (top right); MSG 0º + IODC (bottom left); and HMWR (bottom right). Trends (Kelvin per decade) and standard deviations are indicated in respective inside labels. Only pixels with at least 30 % valid data per month are considered.

The analysis of maximum and minimum LST anomaly differences further supports these findings. Maximum LST anomalies (Figure A2) exhibit the largest deviations, reflecting the greater sensitivity of daytime surface temperatures to surface properties and atmospheric conditions. For maximum LST, anomaly difference trends are consistently larger in magnitude for V_{PREV} than for V3.0 over all GEO domains, including the global domain (-0.99 versus -0.44 K per decade), GOES-16 (0.38 versus 0.03 K per decade), MSG 0º + IODC (-1.54 versus -0.28 K per decade), and HMWR (-1.53 versus -1.08 K per decade). In contrast, minimum LST anomaly differences (Figure A3) show smaller discrepancies between V3.0 and V_{PREV} in respect to ERA5, with a general similar temporal behaviour. For the global domain, despite the overall lower variability with V3.0 compared to V_{PREV} (0.10 versus 0.14 K), the corresponding anomaly differences trend is larger in magnitude for the former (-0.08 versus -0.04 K per decade). This behaviour is related to differences in sampling at the global scale, since V3.0 retains a larger number of observations, and to the fact that minimum temperatures mostly

represent nighttime conditions, where weaker anomaly signals are more susceptible to small variations in sampling.

These results confirm that the implemented input updates in the reprocessed version V3.0 lead to a more temporally consistent representation of both daytime and nighttime LSTs, with the largest added-value observed for daytime-related metrics (i.e. LST max). It is worth noting that, according to the CLMS LST product requirements [GCOS#245], at the global domain and across the individual GEO domains, the mean LST stability achieved in V3.0 meets the base requirement of 0.2 K per decade, while the goal requirement of 0.1 K per decade is achieved for the GOES domain. Trend magnitudes remain below the threshold of 0.3 K per decade in all cases. In contrast, V_{PREV} shows trend magnitudes that are closer to, or in some cases exceed, the target thresholds, indicating lower temporal consistency.

5.2.3 Temporal stability with Sentinel-3A

LST monthly anomaly differences (Figure 13, top panel) are derived by removing the mean annual cycle from each monthly mean, highlighting distinct behaviours for daytime and nighttime conditions. During daytime, anomaly differences are generally positive until the beginning of 2022, reaching a peak of about +0.3 K in September 2018, and become predominantly negative thereafter, with values down to about -0.2 K after April 2023. This pattern is associated with a negative trend of -0.45 K per decade and a variability (i.e. total standard deviation) of 0.12 K, which is comparable to the global LST maximum anomaly differences with ERA5 over 2018–2025 (Figure A2, Appendix), where a similar trend (-0.44 K per decade) and a higher variability (0.24 K) are observed. For nighttime, anomaly differences show a more stable behaviour throughout the study period, with a weak positive trend of +0.11 K per decade and a smaller total standard deviation of 0.09 K, indicating higher temporal consistency between V3.0 and SLSTR-A during nighttime conditions. The corresponding fraction of valid matchup pixels (Figure 13, bottom panel) is typically low (3 to 5 %) due to the requirement that both datasets must contain at least 30 % valid observations per pixel. Higher valid-pixel fractions are observed for nighttime matchups, consistent with generally lower nighttime cloud cover.

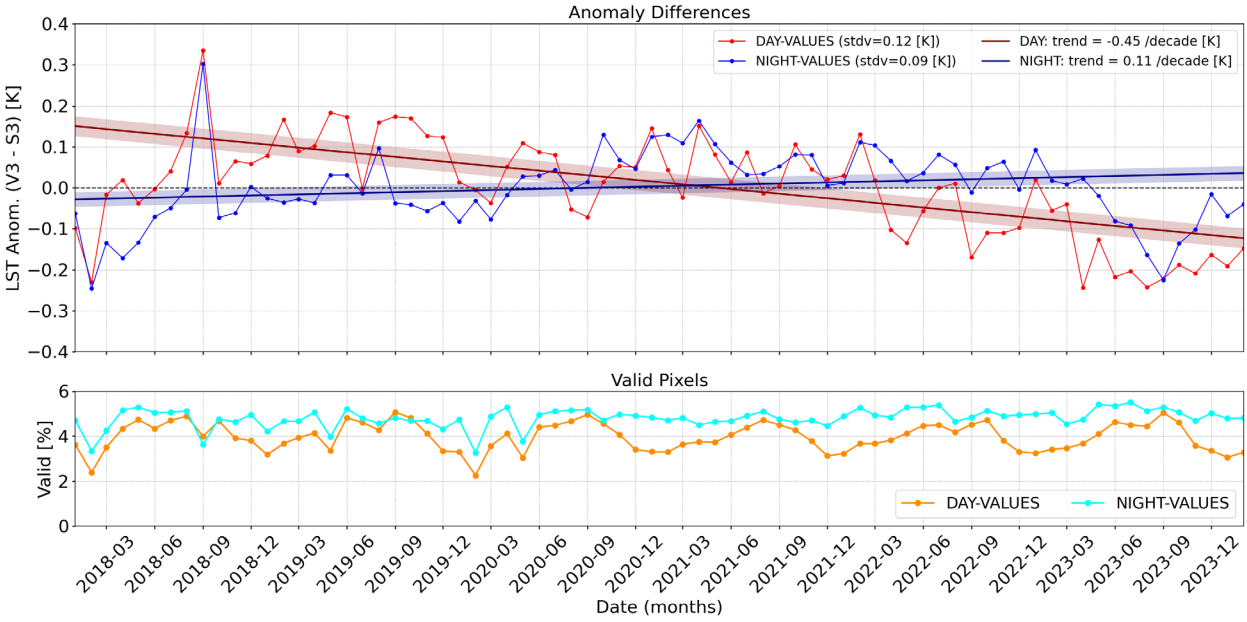


Figure 13 - Monthly anomaly differences (in Kelvin) between CLMS LST V3.0 and SLSTR-A LST for daytime and nighttime values (top), including corresponding linear trends (Kelvin per decade) and standard deviations (stdv) from January 2018 to February 2024. Associated valid pixels (bottom) are shown, where only pixels with at least 30 % valid observations per month are considered.

5.3 ANOMALY CASE STUDIES

The CLMS 10-daily synthesis products offer an additional tool for monitoring temperature anomalies and assessing thermal stress. In particular, the LST10-TCI highlights areas affected by heatwaves, characterized by elevated thermal stress values. While these do not present a quantitative validation of the LST10-TCI product – which is not possible, they provide an event-based qualitative assessment of the product's spatial and temporal consistency with temperature and vegetation anomalies.

Two examples of heatwave events are presented, considering only pixels with at least 30 % valid observations. A climatology for the period 2018–2025 (excluding the event year) is used as the reference baseline. Similarly to the section 5.1.4, the results are shown only for the LST10-Syn median product, since it provides the most representative indicator of surface thermal stress among the synthesis products, being less sensitive to extreme values than the maximum and minimum.

Figure 14 illustrates the evolution of the 2022 summer heatwave over Europe during the first (left panels) and second (right panels) dekads of July. The top row shows the LST10-Syn median anomalies, the middle row the TCI, and the bottom row the CLMS FCOVERv2 anomalies, which provide an independent reference for vegetation conditions. By incorporating the CLMS FCOVERv2 product [CGLOPS1_ATBD_FCOVER300m-V2], we can assess how thermal stress relates to surface

vegetation variability, offering a synergetic perspective that enhances the evaluation of our 10-daily LST products. A clear intensification of temperature anomaly is observed from the first to the second dekad of July, particularly over western Europe, including the Iberian Peninsula and, most notably, France. Anomalies exceeding 4 K correspond well with high thermal stress values (TCI null-values) in the same regions, as well as with reductions in the vegetation signal (i.e. FCOVERv2 negative anomaly values). These patterns are in agreement with the findings of Martins et al. (2024), confirming that the 2022 summer heatwave resulted in both strong thermal stress and pronounced vegetation responses across affected areas.

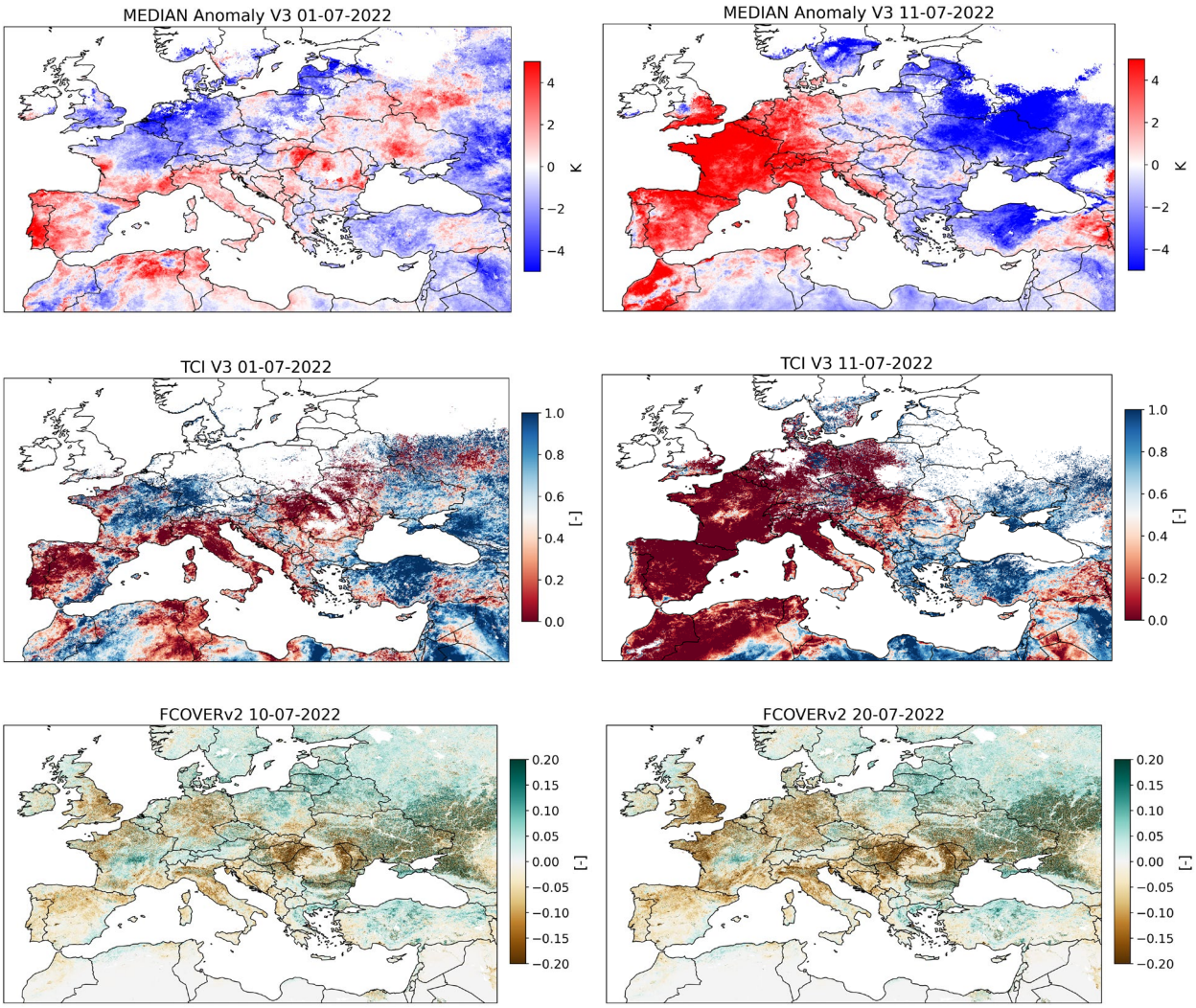


Figure 14 – V3.0 LST10-Syn median (in Kelvin) anomalies, TCI, and CLMS FCOVERv2 anomalies for the first (left) and second (right) dekads of July 2022. Anomalies are computed using the 2018–2025 period, excluding 2022, as reference baseline.

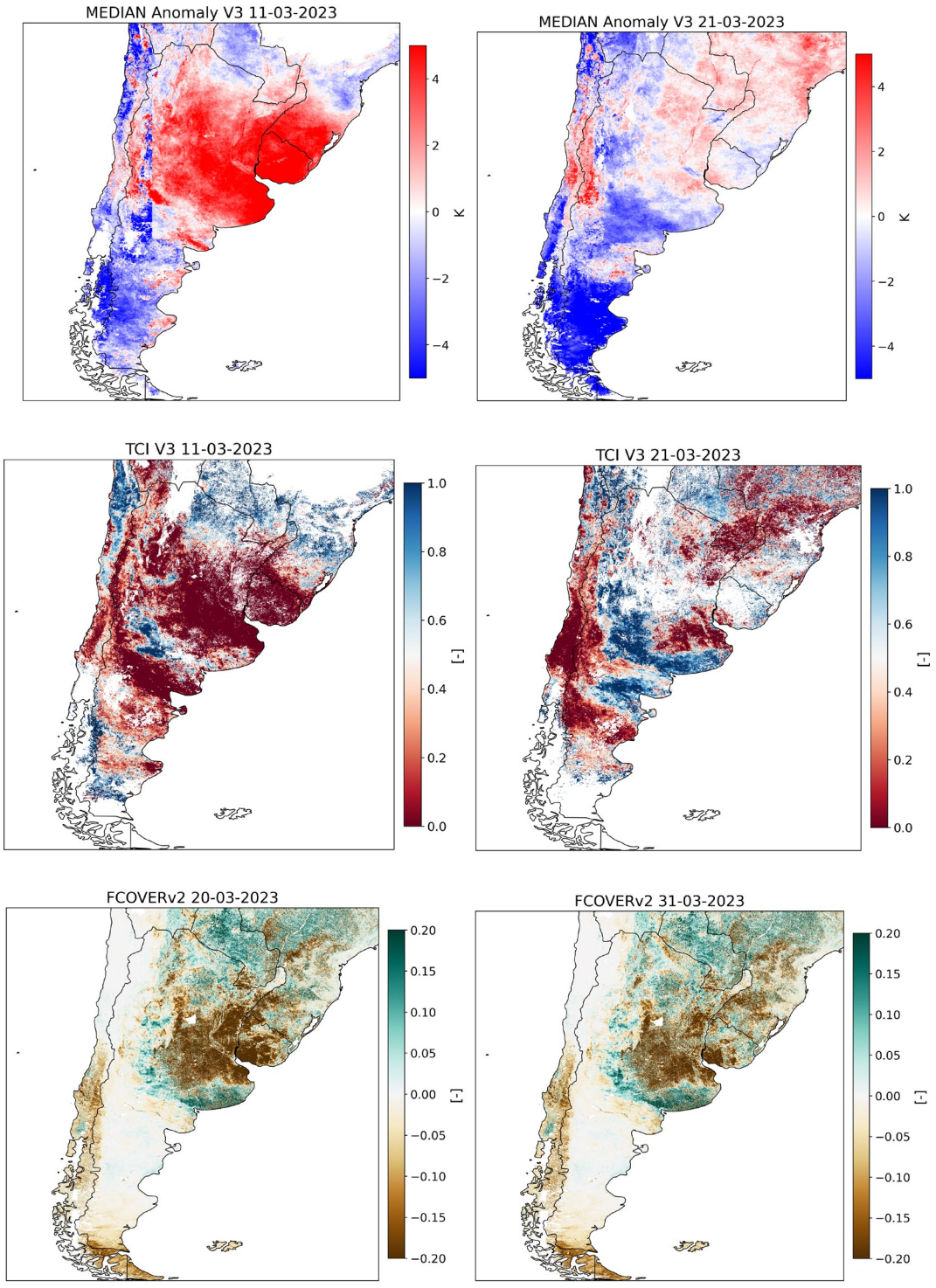


Figure 15 – Same as Figure 14 but for the 2023 southeastern South America heatwave considering the second (left) and last (right) dekads of March. Anomalies are computed using the 2018–2025 period, excluding 2023, as reference baseline.

Other heatwave event example occurred in southeastern South America, affecting the central-eastern Argentina and Uruguay during the late-summer period from the end of February to mid-March 2023, as described by Kim et al. (2025). In Figure 15, the second dekad of March (left panels) shows pronounced positive anomalies (exceeding +4 K) over Uruguay and parts of northern Argentina. These strong anomalies are consistent with the TCI signal, which is predominantly saturated with null-values (TCI = 0), indicating severe thermal stress conditions. In the last dekad of March (right panels), the distribution of positive anomalies is substantially reduced, with values generally ranging between -1 and +1 K over the affected areas, highlighting the weakening of the event. A similar behaviour is observed in the TCI signal, with values increasing to about 0.5 over most of Uruguay, indicating a return to more moderate thermal conditions. The FCOVERv2 anomalies show a spatial pattern consistent with the LST and TCI signals, supporting the observed evolution and spatial footprint of the heatwave event.

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6 CONCLUSIONS AND RECOMMENDATIONS

The main goal of the CLMS LST product is to supply its users with long term global time-series of Land Surface Temperature. Its most distinct feature compared to other global LST products is the fine sampling of the diurnal cycle. The CLMS LST product relies on a series of measurements made by Imagers onboard GEO satellites operated by different agencies (NOAA, EUMETSAT, and JMA). Each agency has its own schedule, and improved satellites have been progressively launched by each agency, namely HMWR-8/9 by JMA and GOES-16/19 by NOAA. New sensors onboard these platforms allow the use of more accurate retrieval algorithms, particularly because they allow higher spectral, temporal and spatial resolutions. This poses challenges to operational products such as the CLMS LST, since retrievals from all satellites used to produce LST estimates should be as harmonized as possible. It is therefore of paramount importance that the algorithms are continuously developed to improve retrievals from each individual sensor, but also to improve the harmonization between estimates from different sensors, especially over areas where measurements from different sensors overlap. It is also desirable that this harmonization extends to the different land surface products provided by the CLMS, which use a common set of inputs for their retrievals, whenever required and/or possible.

With the abovementioned aspects in mind, a new version (V3.0) of the CLMS LST product has been prepared. The changes in this version include the spatial resolution upgrade from 5 to 2.5 km on a regular latitude–longitude grid with 0.025º spacing, enhancing the representation of local surface heterogeneity and enabling a more harmonized product by using data from all GEO platforms closer to their native resolutions. The full integration of the most recent NWC-SAF cloud information for GOES and HMWR platforms improves the reliability of LST retrievals. This update reduces the probability of cloud-contaminated pixels being included in the LST estimates. Furthermore, the NWC-SAF cloud masking brings the GOES and HMWR retrievals into closer alignment with the ones from MSG 0° and IODC data, which have already employed the same cloud detection software. Thereby, further cross-platform harmonization is enhanced and more consistent global LST fields are ensured. Another important improvement concerns the refined snow cover treatment. It now integrates the CLMS SCE 1 km product with the IMS 4 km product dataset for gap-filling over the Northern Hemisphere. This method incorporates a persistence-based gap-filling approach that enables a more complete and temporally consistent representation of snow conditions. As a result, unrealistic fluctuations in surface emissivity are reduced, leading to more accurate characterizations of snow-affected surfaces and, consequently, their associated LST estimates. The CLMS LST V3.0 also incorporates a comprehensive refactoring of the LST retrieval algorithm, aimed to enhance the overall computational efficiency and to ensure consistency over all GEO platforms datasets. This implementation not only improves the robustness of the operational processing chain but also facilitates NRT updates and maintenance activities moving forward. Furthermore, a full historical reprocessing has been carried out for the period 2018–2025 with the aim of providing temporal

consistency over the entire dataset and removing discontinuities introduced by incremental updates in earlier operational versions.

6.1 CONCLUSIONS

In this validation report, an extensive quality assessment of the CLMS LST V3.0 product was performed, extending the standard procedure typically adopted in routine validation exercises of the CLMS LST operational product. The V3.0 was shown to provide a globally consistent, temporally stable, and spatially detailed LST dataset, offering more realistic and reliable representation of surface temperatures in comparison to the previous operational version (V_{PREV}). The comparison of LST estimates generated from each product version against independent measurements is essential to assess the impact and added value of the implemented updates. Comparisons to in-situ LST observations included a total of 37 stations from the GBOV network representing a wide range of land cover types and climate conditions. For each station, both V3.0 and V_{PREV} datasets spanning from January 2018 to December 2024 were matched against corresponding in-situ estimates using the satellite's central pixel nearest to each site location. Comparisons revealed overall better scores in the statistical performance with V3.0, namely when considering same hourly matchups between V3.0 and V_{PREV} for the median difference (-0.43 K versus -0.59 K) and the median absolute residuals (1.76 K versus 1.81 K), while comparable RMSDs were found (3.24 K versus 3.23 K). It is worth noting that these results are station-dependent, with many of the stations located in, e.g. regions with seasonally varying vegetation/snow cover conditions. The conformity tests results further reinforced the validation findings. Overall, for the new V3.0 version, 53 % of satellite/in-situ matchups meet the CLMS product's threshold requirement (absolute error $\leq$ 2 K), while 29 % meet the goal requirement (absolute error $\leq$ 1 K). These values were found to be higher than those from V_{PREV} (51 and 27 %, respectively), indicating an increased compliance with the product specifications with V3.0, and reflecting the robustness and consistency of the reprocessed LST estimates under diverse environmental conditions. The direct validation was further extended to the CLMS 10-daily products derived from the hourly dataset, including the LST10 daily-cycle (LST10-DC) and synthesis (LST10-Syn) products. Generally, the results showed that V3.0 provides consistent performance with improvements relative to V_{PREV}, particularly for the maximum and median temperatures.

The second part was focused on the spatial consistency and temporal stability of the product. When cross-compared with V_{PREV} for two separate periods (2018–2020 and 2021–2025), V3.0 shows improved spatial consistency. This is particularly observable over the MSGs domain due to the incorporation of the IODC data for the full reprocessing period, while this component is present in V_{PREV} only after January 2021. The first temporal stability assessment consisted in comparing CLMS LST V3.0 against ERA5 skin temperature reanalysis over the full 2018–2025 period. Anomaly differences derived from the monthly means were used following the removal of the mean seasonal cycle. This approach allowed a direct assessment of temporal stability and consistency between

product versions. On the global scale, V3.0 anomaly differences for the mean LST values remained generally within ±0.2 K until the end of 2022, followed by a short period of larger negative anomalies during the second half of 2023 associated to the sensor transition from MSG4 to MSG3, and a gradual decrease towards near-neutral anomalies in 2025. The increased stability of V3.0 is reflected in its weaker trend (-0.15 K per decade) and lower variability (total standard deviation of 0.12 K) in comparison to V_{PREV} (-0.25 K per decade and 0.16 K, respectively). Over each individual GEO domain, V3.0 systematically showed reduced anomaly differences together with lower trends and variability, reinforcing its overall temporal consistency. The implementation of dynamic vegetation-based surface emissivity and, to a lesser extent, snow cover-based emissivity updates over the whole time period allowed to address the temporal discontinuities previously observed in V_{PREV}, particularly over the GOES-16 and HMWR components. Further temporal assessment was performed by comparing CLMS LST V3.0 estimates with the ones from an independent LST retrieval system, namely the SLSTR sensor onboard Sentinel-3A (SLSTR-A). The anomaly-based assessment of the monthly means confirmed the temporal stability of the reprocessed version. Daytime anomalies are generally positive before 2022 (with a peak around +0.3 K in September 2018) and predominantly negative thereafter (down to about -0.2 K), with a corresponding trend of about -0.45 K per decade and a total standard deviation of 0.12 K. Nighttime anomalies exhibit a more stable behaviour, with a weak positive trend of about +0.11 K per decade and a total standard deviation of 0.09 K. Overall, the anomaly-based assessments demonstrated that V3.0 provides a more coherent temporal representation of LST in relation to other global products.

The last section of this report was dedicated to application examples of the synthesis products to show their added value in detecting heatwave signals. Two case studies, corresponding to the European summer 2022 and the late-summer 2023 over southeastern South America showed that the combination of LST10-Syn median anomalies with the Thermal Condition Index (TCI) effectively captures the spatial distribution and evolution of thermal stress. This analysis was complemented with CLMS FCOVERv2 anomalies, which provide an independent indicator of vegetation conditions and highlight the benefits of combining CLMS land surface products for monitoring extreme temperature events.

In conclusion, CLMS LST V3.0 provides users with a seamless, harmonized LST time-series suitable for both climate and environmental monitoring applications.

6.2 RECOMMENDATIONS

Following the main conclusions and results presented above, several aspects can be considered to further improve the CLMS LST products, their monitoring, validation and use.

Future product updates should focus on integrating Meteosat Third Generation (MTG) observations, which provide 2 km nadir resolution, ensuring compatibility and harmonization with existing GEO

datasets. Continued use of high-resolution dynamic vegetation, and enhanced cloud masking will enhance the temporal stability, spatial consistency, and overall accuracy of the CLMS LST product.

In-situ observations are often not fully representative of LST at pixel scale. The deployment of additional stations would be valuable, provided their locations are carefully selected and are well-maintained to ensure surface representativeness at the satellite pixel scale.

Finally, for the LST10 diurnal cycle and synthesis products, we recommend users to apply a threshold value of 0.3 on the fraction of valid data to avoid sampling problem. Moreover, the maximum and minimum fields on the LST10 Synthesis product should be used with caution due to sampling uncertainty, as we recommend to use the median instead.

7 APPENDIX

The data availability for both operational and reprocessed (V3.0) global LST versions, together with each GEO component used in the production of the CLMS LST global product, is shown in Figure A1. While the reprocessed version (V3.0) exhibits 100 % availability over the entire reprocessing period (2018–2025), the current operational product (GEOs) shows data gaps that are intrinsically related to the availability of the individual contributing missions.

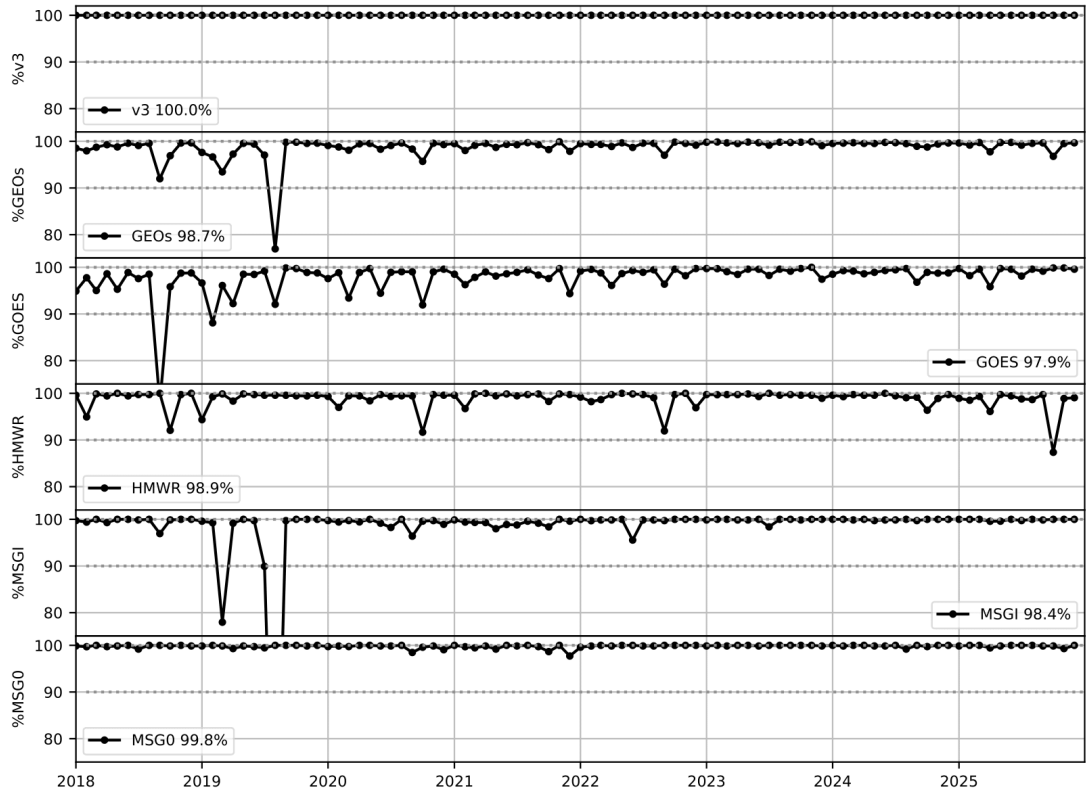


Figure A1 - Data availability (in %) used for the production of the CLMS LST product, including the operational global LST (GEO-based) and the reprocessed V3.0 versions, as well as the individual GEO components: GOES-16, HMWR, MSG-IODC, and MSG 0º. The corresponding mean availability is indicated in each label.

As complement to the monthly mean anomaly differences between the CLMS LST (V3.0 and V_{PREV}) product and ERA5 skin temperature (section 5.2.2), the following figures comprise the same outputs but considering the maximum (Figure A2) and minimum (Figure A3) LST values.

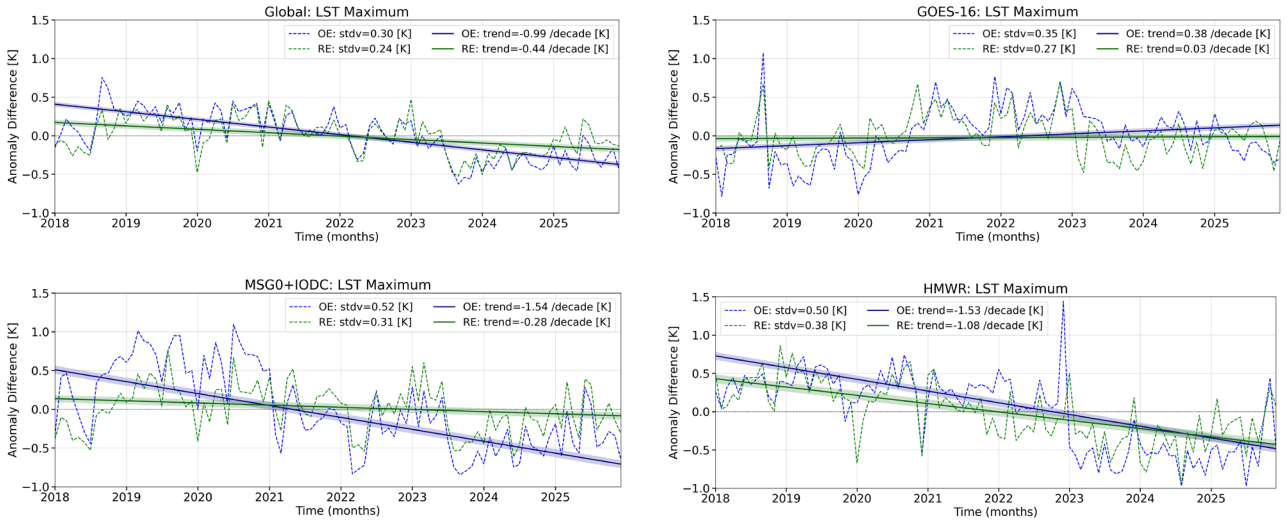


Figure A2 – Same as Figure 12 but for the maximum LST values.

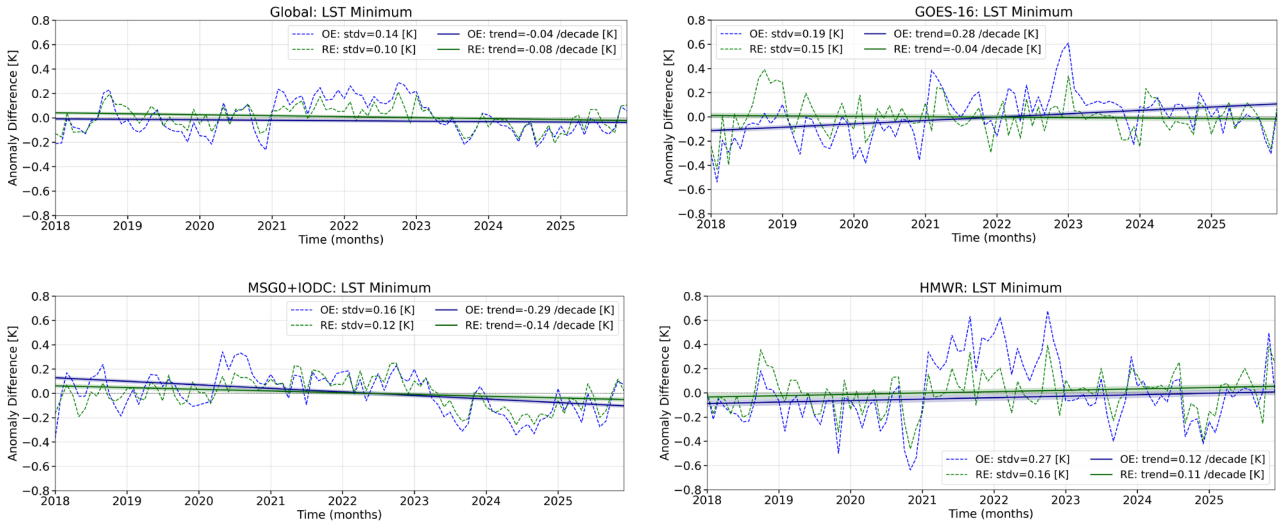


Figure A3 – Same as Figure 12 but for the minimum LST values.

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